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# Induced Paths, Radius, and Extremal Neighborhood Independence in Graphs without 4-Cycles

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**Abstract**

Let $M(n)$ be the maximum of $\ell(G) = S(G)/n$ over connected graphs of order $n$ with no 4-cycle subgraph and no induced path on six vertices, where $S(G)$ is the sum of $\alpha(G[N(v)])$. Then $M(7) = 16/7$, $M(8) = 5/2$, $M(9) = 8/3$, $M(10) = 3$ and $M(n) = 2 + 2/n$ for every $n \geq 11$. For every $n \geq 7$, the extremal graph is unique up to isomorphism. The maximum is the value 3, attained only by the Petersen graph. Deleting one vertex of the Petersen graph gives the unique extremal graph of order 9, and a single graph $H_8$ is extremal at order 8. For every $n \geq 11$ the unique extremal graph consists of a 6-cycle, one further vertex joined to an antipodal pair, and $n - 7$ leaves at that vertex. In particular $S \leq 3n$, with equality only for the Petersen graph.

The extremal family changes once the order exceeds 10. If an induced 6-cycle meets at least two antipodal pairs in off-cycle common neighbours, the second distance layer contains at most one vertex, and the order is at most 10. At order 11 that branch is empty, and the unique extremal graph is obtained from the unique extremal graph of order 7 by placing the remaining vertices as leaves at the designated off-cycle vertex. The induced 5-cycle branch contributes at most $2n + 1$ to $S$, and the chordal branch at most $2n - 2$.

A connected graph with no 4-cycle subgraph, of radius 4 and induced-path order 7, is either $C_8$ or is obtained from a 6-cycle by a matching-and-isolate extension at each cycle vertex, together with at most one additional vertex on each cycle edge. For every such graph $S = 2n - b - m$, and $S = 2n$ precisely for $C_8$ and for the graphs obtained from $C_6$ by attaching at least one leaf at every cycle vertex.

Every connected triangle-free graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $2e \leq 3n$, with equality only for the Petersen graph.

These statements do not use Conjecture 133 of *Written on the Wall II*. As an application, the theorem of Erdős, Saks and Sós yields that conjecture, in the real-valued form $p(G) \geq r(G) + \ell(G)$, whenever $\ell \leq 3$, whenever $r \leq 2$ and $\ell \leq 4$, whenever $r = 4$ and $\ell \leq 4$, and whenever $r \geq 5$ and $\ell \leq 4$. Every remaining counterexample with $\ell \leq 4$ has order at least 14, contains at least one triangle, and lies in the cell $r = 3$, $p = 6$ and $3 < \ell \leq 4 - 13/n$. The value $\ell = 4$ does not occur for a connected graph with no 4-cycle subgraph and no induced path on seven vertices. The conjecture itself remains open.

## 1 | Introduction

All graphs are finite and simple. For a graph $G$ with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where $\alpha$ is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if $G$ contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong $C_4$-free*.

The two structural results of the paper are Theorems 3.6 and 4.3. Let $P$ denote the Petersen graph and $P - v$ the graph obtained by deleting one vertex. Theorem 3.6 determines the maximum $M(n)$ of $\ell$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$. The value is $16/7$ at order 7, $5/2$ at order 8, attained only by the graph $H_8$ of Figure 1, $8/3$ at order 9, attained only by $P - v$, and 3 at order 10, attained only by $P$. For every $n \geq 11$ the maximum is $2 + 2/n$. For every $n \geq 7$, the extremal graph is unique up to isomorphism. The same theorem gives $S \leq 3n$, with equality only for $P$.

The shape of the extremal graph changes between orders 10 and 11, and the change is forced by the induced 6-cycle branch. A vertex off such a cycle meets it in an antipodal pair. Distinct off-cycle vertices of the first distance layer use distinct pairs, so there are at most three of them. If at least two antipodal pairs are occupied, any two vertices of the second distance layer would have two common neighbours in the first layer. That layer therefore has size at most one, and the order is $6 + k + h \leq 10$. The graph of order 10 in this branch is $P$. At order 11 the branch with two or more occupied pairs is empty. The only remaining graphs in the induced 6-cycle branch are those with a single off-cycle vertex $a$ joined to one antipodal pair, together with leaves at $a$: the graph $\Gamma_0$ of Figure 1 with $n-7$ leaves attached at $a$. This graph has $S = 2n+2$. Every graph in the induced 5-cycle branch has $S \leq 2n + 1$, and every chordal graph in the class has $S \leq 2n - 2$, so both fall short of $2n + 2$. Uniqueness can fail for smaller orders. The same case division shows that at order 4 both $P_4$ and $K_{1,3}$ attain $S = 6$, and that at order 6 both $C_6$ and the graph obtained from $C_5$ by adding one leaf attain $S = 12$.

An enumeration of the connected strong $C_4$-free $P_6$-free graphs of order at most 12 agrees with this uniqueness statement. It is not used in the proofs.

| $n$ | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|
| number of graphs | 43 | 87 | 183 | 355 | 699 | 1327 |
| maximum of $S$ | 16 | 20 | 24 | 30 | 24 | 26 |
| number of extremal graphs | 1 | 1 | 1 | 1 | 1 | 1 |

Theorem 4.3 describes every connected strong $C_4$-free graph of radius 4 and induced-path order 7. Fajtlowicz's *r*-ciliates, as summarised by DeLaViña, Larson, Pepper and Waller [8], already supply an induced skeleton in a graph of radius 4: an even cycle with a pendant path of one common length at every vertex. The prohibition on 4-cycle subgraphs, together with the absence of an induced $P_8$, rules out the two ciliate skeletons whose core contains a 4-cycle and leaves $C_8$ or the 6-cycle with a pendant leaf at each vertex. The new content is the classification of every extension of that skeleton. The sets $C_i$ of vertices with trace $\{v_i\}$ may be nonempty simultaneously, the only edges off the cycle are a matching inside a single $C_i$ or a vertex glued to one cycle edge, and $S = 2n - b - m$, where $b$ is the number of glued vertices and $m$ is the number of matching edges. The device that forbids an edge with ends in distinct sets $C_i$ is a distance-4 witness $w_i \in C_i$: such an edge completes an induced $P_8$ through two of these witnesses. Figure 2 draws that path, and on the same cycle it draws an isolated leaf, a matching edge and a glued vertex, so that the contributions to $b + m$ are visible. Corollaries 4.4 and 4.5 give $S$ and the equality case. Neither theorem is a special case of the conjecture recorded next.

**Conjecture 1.1** (Graffiti/WOWII Conjecture 133)**.** *Every connected graph $G$ on at least two vertices satisfies*

$$r(G) + \ell(G)^{c_{C_4}(G)} \leq p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [7]. We follow the formal statement in the Formal Conjectures repository at the fixed commit 5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e [10], the merge commit of pull request #5727 [11]. That commit removed the floor function from an earlier version of the file. The statement there reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. We work with this real-valued statement and call it REAL-133.

If $G$ contains a 4-cycle, the conjecture reads $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced, so the inequality holds. Only strong $C_4$-free graphs remain. Since $p$ and $r$ are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \;\geq\; \lceil \ell(G) \rceil. \tag{1.1}$$

Theorem 5.11 is the range in which the estimates of this paper give REAL-133. The triangle-free bound is Theorem 5.4. With the deficit lemmas of Section 5, every counterexample in the range $\ell \leq 4$ falls in the cell of Corollary 5.9: radius 3, induced-path order 6, order at least 14, at least one triangle, and $3 < \ell \leq 4 - 13/n$.

H. Chen [3] studies the floor form $p \geq r + \lfloor \ell \rfloor$ of the same conjecture. The arguments below do not use his proofs. His Lemma 2.1 is the local matching lemma: in a strong $C_4$-free graph the neighbourhood of every vertex is a disjoint union of edges and isolated vertices, and $S = 2e - 3T$. Lemma 2.1 records the same description. His Lemma 7.18 states that if an induced 6-cycle lies in a strong $C_4$-free $P_6$-free graph, then every vertex outside the cycle has either no neighbour on the cycle or exactly two, and those two are antipodal. His Corollary 7.20 records the consequence for the cycle vertices: each

has triangle count zero, and its degree is 2 or 3 according as its antipodal slot is empty or occupied. The same restriction on traces is Lemma 3.4 when the cycle dominates, and it is the description of the first distance layer in the proof of Theorem 3.6. His Theorem 6.7 states that a connected $C_4$-subgraph-free graph of diameter 2 with $\ell > 3$ has an induced path on at least six vertices. On the slice $r = 2$ and $3 < \ell \leq 4$ this is the same numerical conclusion as Theorem 5.11. The increment here is that Theorem 5.11 also covers radius 2 when the diameter is larger than 2, while Theorem 3.6 determines the extremal value at every order. Immediately after the proof of his Theorem 3.6, Chen records that every regular graph of girth at least 5 and radius 2 satisfies the floor form, and that this includes every Moore graph of diameter 2. A Moore graph of diameter 2 has $\ell$ equal to its degree, an integer, so the floor form and the real form coincide. The Moore assertion of Proposition 8.4 is an alternate proof of that known case.

The exact function $M(n)$ is not a consequence of those lemmas. The proof of Theorem 3.6 continues to the second distance layer, and it treats the induced 5-cycle branch and the chordal branch separately. For every $n \geq 7$ it also gives uniqueness. Below that range the same division produces two extremal graphs at orders 4 and 6, as recorded above. Theorem 5.4 is the two-step argument from the classification, due to Chen and Xu [5, Lemma 3.4], of the connected triangle-free strong $C_4$-free $P_7$-free graphs with no clique cutset. If $v$ is a cut vertex, induced paths that start at $v$ in distinct components of $G - v$ concatenate, so one side has rooted path order at most 3 and falls under Lemma 5.3. If there is no cut vertex but $\{u, v\}$ is a clique cutset, a shortest $u$–$v$ path in each of two components has three internal vertices, and those six vertices together with $v$ induce a $P_7$. Restoring clique cutsets therefore yields no new equality graph: every graph with a clique cutset is strictly below $2e = 3n$, and the only equality graph on the Chen–Xu list is the Petersen graph.

Examples 9.1 and 9.7 start from an $r$-ciliate of larger radius and add further blocks. They are not part of the radius-4 classification. In Example 9.1 each added path of length four contributes 3 vertices and 4 edges, so it increases $S = 2e$ by 8 while the radius and the induced-path order remain those of the ciliate. In Example 9.7 each attached copy of the incidence graph of a projective plane adds $(q + 1)f$ to $S$ and $f - 1$ to the order, and the radius and the path order remain $t + 3$ and $2t + 5$.

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves the triangle-free bound and the strengthened residual cell. Section 6 is the application to REAL-133, including the cells in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$. The rooted inequalities that remain are in Sections 7 and 8; they are not used for Theorems 3.6, 4.3 and 5.4 or for Corollary 5.9. Table 1 is the list of what is proved and what is open. A finite search is not a proof in this article. Where the repository does not contain a versioned checker, an input list, and a completeness argument, the corresponding statement is not used as a hypothesis.

## 2 | Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing $v$. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

**Lemma 2.1.** *Let $G$ be strong $C_4$-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every $v$, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

*Proof.* (1) Two common neighbours $w, w'$ of $x, y$ give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with $v$. So the components are single vertices and edges, each edge being a triangle at $v$. (3) Sum $d(v) - t(v)$ over $v$. □

The components of $G[N(v)]$ are the *units* of $v$. For a vertex set $U$, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of $U$. We write $L(v) = D_G(\{v\})$.

**Lemma 2.2.** *Let $G$ be strong $C_4$-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

*In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.*

*Proof.* For $x \notin U$, deleting $U$ removes exactly the units of $x$ that lie inside $U$; a unit that is an edge with one end in $U$ survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of $v$ in triangles with $v$ keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

**Lemma 2.3.** *Let $v$ be a cut vertex of $G$. Split the components of $G - v$ into two nonempty groups, and let $G_1, G_2$ be the subgraphs induced by the two groups together with $v$. Then $S(G) = S(G_1) + S(G_2)$.*

*Proof.* $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

**Theorem 2.4** (Erdős, Saks and Sós [9])**.** *Every connected graph $G$ has an induced path on at least $2r(G) - 1$ vertices.*

**Theorem 2.5** (Camby and Schaudt [2, Theorem 5])**.** *Let $k \geq 4$ and let $G$ be a connected graph with no induced path on $k$ vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle $C_k$.*

**Theorem 2.6** (Seinsche [13])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [2, Theorem 5] applied with $H = G$. We write $P_k$ for the path on $k$ vertices, and "$G$ is $P_k$-free" means that $G$ has no induced $P_k$, that is, $p(G) \leq k-1$. A connected graph of radius $r$ has at least $2r$ vertices.

**Lemma 2.7.** *Let $H$ be a connected strong $C_4$-free $P_4$-free graph on at least two vertices. Then $H$ has a vertex adjacent to all other vertices.*

*Proof.* By Theorem 2.6, $H$ is the join of two nonempty vertex sets $A$ and $B$. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

## 3 | Small-path mass bounds

**Lemma 3.1.** *If $G$ is connected and strong $C_4$-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where $m$ is the number of edges of $G[N(u)]$ for a dominating vertex $u$. Hence $\ell(G) < 2$, and (1.1) holds.*

*Proof.* Let $v \neq u$. By Lemma 2.1(1), $v$ has at most one neighbour other than $u$, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \leq 2$. If $G$ is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with $u$ form an induced $P_3$, so $p \geq 3 \geq r + \lceil \ell \rceil$. □

**Lemma 3.2.** *If $G$ is connected and strong $C_4$-free with $r(G) \geq 2$, then $p(G) \geq 4$.*

*Proof.* If $G$ were $P_4$-free, Lemma 2.7 would give a dominating vertex, so $r(G) \leq 1$. □

**Lemma 3.3.** *If $G$ is connected, strong $C_4$-free and $P_5$-free, then $S(G) \leq 2n - h$, where $h$ is the number of vertices $v$ with $\lambda(v) \geq 3$. In particular $S(G) \leq 2n$.*

*Proof.* Call $v$ *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \geq 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of $x$ if $x$ lies in an edge unit of $v$.
*Key fact.* Let $x, y$ lie in different units of $v$. A common vertex of $X_x$ and $X_y$ would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced $P_5$: $x \nsim y$ because they lie in different units, $x \nsim y'$ and $y \nsim x'$ by disjointness, and $v$ is adjacent to neither $x'$ nor $y'$. So $X_x$ is complete to $X_y$. If both are nonempty and $|X_x| \geq 2$, a vertex of $X_y$ has two neighbours in $X_x$, which are also neighbours of $x$, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.
*Main step.* For each high $v$ we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is $v$, one in each of $\lambda(v) - 1$ distinct units of $v$.
Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every $y$ outside the unit of $u$. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so $y$ is simplicial; so is $\bar{y}$. Hence $v$ is the only high neighbour of $y$, and the $\lambda(v) - 1$ units other than that of $u$ supply the vertices.
Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Then $\lambda(x) \leq 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices $x$ has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let $x$ lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced $P_5$: $x \nsim y$ (different units), $x \nsim y', z'$ (disjointness), $v \nsim y', z'$, and $y \nsim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is $v$. This gives at least $\lambda(v) - 1$ units.

*Counting.* The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}}(\lambda(v)-1) \le \#\{\text{simplicial vertices}\}$, and

$$S = \sum_{v} \lambda(v) \le 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v)-2) \le 2n - h. \qquad \square$$

**Lemma 3.4.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free with a dominating induced cycle $C = v_0v_1\cdots v_5$, and let $W = V(G)\setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of $C$, distinct vertices of $W$ use distinct pairs, and $W$ is independent. Hence $|W| \le 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \le 2$ if $W \ne \emptyset$.*

*Proof.* Indices are modulo 6. Let $w \in W$; it has a neighbour on $C$. If $w$ has exactly one neighbour $v_i$, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced $P_6$. If $N(w)\cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced $P_6$. Two neighbours at distance 2 on $C$ give a 4-cycle. A set of at least three vertices of $C_6$ contains two vertices at distance 2. So $N(w)\cap C$ is an antipodal pair. Two vertices of $W$ with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on $C$, and $wv_iv_jw'$ is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if $w$ uses $\{v_i, v_{i+3}\}$, then every vertex of $C$ is within distance 2 of $v_i$ (the vertex $v_{i+3}$ through $w$), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So $v_i$ has eccentricity 2. $\square$

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong $C_4$-free $P_6$-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form.

**Lemma 3.5.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free. Then either $G$ has a vertex of eccentricity at most 2, or $G = C_6$.*

*Proof.* Apply Theorem 2.5 with $k = 6$ to get a connected dominating set $X$. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is $P_4$-free. If $|X| = 1$, the vertex in $X$ dominates $G$. If $|X| \ge 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since $X$ dominates $G$, every vertex is within distance 2 of $u$. $\square$

**Theorem 3.6.** *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong $C_4$-free $P_6$-free graphs of order $n \ge 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \ge 11. \end{cases}$$

*For every $n \ge 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph $P$. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph $H_8$ of Figure 1. For $n = 7$ and for every $n \ge 11$ it is the graph $\Gamma_{n-7}$ of that figure: a 6-cycle, a vertex $a$ adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to $a$. In particular, every connected strong $C_4$-free $P_6$-free graph $G$ of order $n \ge 2$ satisfies $S(G) \le 3n$, with equality if and only if $G$ is the Petersen graph. At orders 4 and 6 the maximum is attained by two nonisomorphic graphs.*

*Proof.* Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced $P_6$. Thus the only possible holes have lengths five and six.

*Case 1: an induced six-cycle.* Label its vertices $c_0, \dots, c_5$, indices modulo 6, and write $A$ for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of $A$ use distinct pairs, and $A$ is an independent set: an edge of $A$ joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \le 3$. Figure 1 shows the three extremal graphs of this branch.

Every vertex $x$ at distance two from the cycle is adjacent to every vertex of $A$. Such an $x$ has at least one neighbour in $A$. If $x \sim a$ and $x \nsim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of $b$ lies at cycle distance two from some contact of $a$, and we orient the cycle from that contact of $a$ toward that contact of $b$. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced $P_6$. Indeed $x$ meets the cycle in nothing, $a$ meets the cycle only in its antipodal pair (neither $c_{i+1}$ nor $c_{i+2}$), $b$ meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \nsim c_i, c_{i+1}$), and $a \nsim b$.

There is no vertex at distance three or more from the cycle. A shortest path $z, x, a, c_i$ from such a vertex down to the cycle, extended by $c_{i+1}, c_{i+2}$, is an induced $P_6$: the vertex $z$ has no neighbour in $A$ or on the cycle, and $x$ has no neighbour on the cycle.

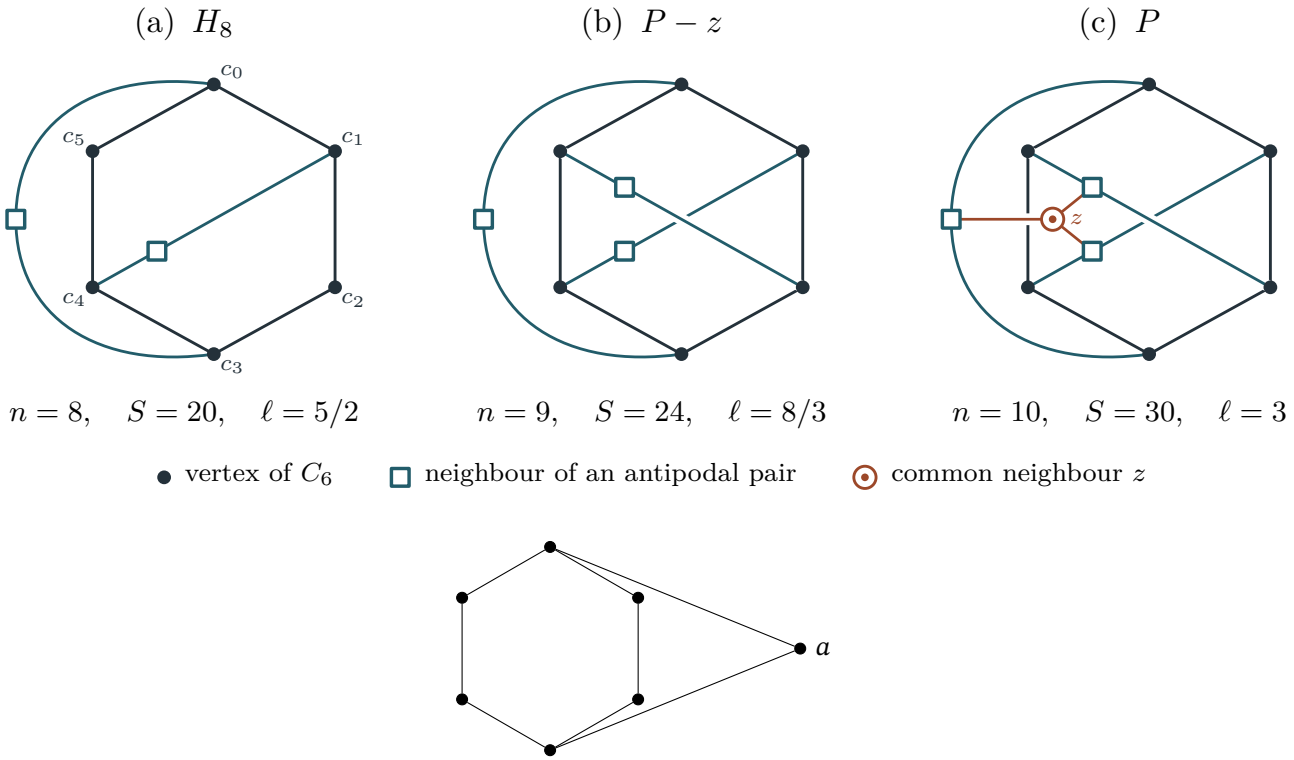



**FIGURE 1** | Extremal graphs for $M(n)$. In $H_8$, two antipodal pairs on the 6-cycle have a common neighbour outside the cycle. Adding a common neighbour for the third pair gives $P-z$. Joining a new vertex $z$ to the three outside vertices gives the Petersen graph $P$. The three graphs are triangle-free, so $S = 2e$. The drawing underneath is $\Gamma_0$, the unique graph of order 7 with $\ell = 16/7$: the marked vertex $a$ is adjacent to one antipodal pair. Attaching any number of leaves at $a$ gives the unique extremal graph of each order $n \geq 11$, with $S = 2n + 2$.

Let $h$ be the number of vertices at distance two. If $k = 0$, then $A$ is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with $a$. Writing $m$ for the number of edges of the matching gives $n = 7+h$, $e = 8+h+m$ and $T = m$, so $S = 16+2h-m = 2n+2-m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that $G$ is the graph $\Gamma_h$ formed by the 6-cycle, the vertex $a$ on one antipodal pair, and $h$ leaves at $a$.

The graph $\Gamma_h$ lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from $a$ to the cycle, and each of the two $a$-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced $P_6$. A leaf, having degree one, occupies an end of any path that contains it, and that path meets $a$ next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in $\Gamma_0$ that starts at $a$. No such path exists: the two neighbours of $a$ on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord $ac_{i+3}$, and every shorter continuation from $a$ through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to $a$. An induced path of $\Gamma_0$ that avoids $a$ lies on the 6-cycle and has order at most five. An induced path of $\Gamma_0$ that contains $a$ in its interior has $c_i$ and $c_{i+3}$ as the two neighbours of $a$ on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus $\Gamma_h$ is strong $C_4$-free and $P_6$-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of $A$. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of $A$ and to nothing on the cycle, $A$ is independent, and each vertex of $A$ contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

| $(k, h)$ | $(2, 0)$ | $(2, 1)$ | $(3, 0)$ | $(3, 1)$ |
|---|---|---|---|---|
| $S/n$ | $5/2$ | $8/3$ | $8/3$ | $3$ |

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write $a_i$ for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices $a_0, a_1, a_2$ and a vertex $z$ adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5.

Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding $a_i$; the vertex $z$ meets every $a_i$; and a non-neighbour of $a_i$ on the cycle is adjacent to $c_i$ or to $c_{i+3}$. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives $v_0$ a common neighbour with each of $v_3, v_4, v_5$. Those neighbours lie off the path, because the only neighbour of $v_0$ on the path is $v_1$ and the path is induced. They are distinct: a vertex adjacent to two of $v_3, v_4, v_5$ would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: $v_0$ has degree 3, so it has only two neighbours off the path. Thus the graph is $P_6$-free. It is the Petersen graph $P$, and $\ell(P) = 3$.
Deleting $z$ leaves $(k, h) = (3, 0)$, the graph $P-z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex $z$, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by $c_5$, $z$ and $c_2$, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of $P$ are strong $C_4$-free and $P_6$-free. The pair $(2, 0)$ is the graph $H_8$ in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither $P$ nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.
*Case 2: an induced five-cycle and no induced six-cycle.* Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from $C$ has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on $C$ and forms a 4-cycle. If a vertex $x$ were at distance two, with a neighbour $a$ in the first layer, the path $x, a$ followed by four consecutive cycle vertices, beginning at one contact of $a$ and omitting the second contact if $a$ has one, would be an induced $P_6$. Thus $C$ dominates $G$. Write $h$ for the number of vertices with singleton trace and $j$ for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside $C$ are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of $C$. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let $m$ be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at $c_i$ meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of $c_i$ and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let $b$ be the number of singleton-to-pair edges, so $b \leq \min(h, j)$.

Consequently $n = 5 + h + j$ and $e = 5 + h + 2j + m + b$. The triangles are the $j$ pair triangles and the $m$ matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer $b$ is at most 1. The opposite edge of the support $c_i$ is $\{c_{i+2}, c_{i+3}\}$, and as $i$ runs through the cycle this edge runs through every edge exactly once. A singleton meets at most one vertex of the opposite pair class, and that class contains at most one vertex. Thus two singleton–pair edges use two distinct supports and the two opposite pair vertices. Supports at distance 1 are nonadjacent, for an edge between the corresponding singletons would form a 4-cycle with the cycle edge; supports at distance 2 are nonadjacent, for an edge between the corresponding singletons would form an induced 6-cycle along the longer arc. Pair vertices on distinct edges are nonadjacent, as already recorded.
Suppose the supports are at distance 2: a singleton $s$ at $c_0$ is adjacent to the pair vertex $p$ on $\{c_2, c_3\}$, and a singleton $t$ at $c_2$ is adjacent to the pair vertex $q$ on $\{c_4, c_0\}$. The path $s, p, c_2, t, q, c_4$ is an induced $P_6$. The consecutive non-edges are the trace restrictions. The distance-2 pairs on the path are $sc_2$, $pt$, $c_2q$ and $tc_4$: $s$ meets only $c_0$ on the cycle, $t$ meets only its opposite pair vertex $q$, $q$ meets the cycle only in $\{c_4, c_0\}$, and $t$ meets the cycle only in $c_2$. The distance-3 pairs are $st$, $pq$ and $c_2c_4$, all non-edges by the preceding paragraph or by the cycle. The distance-4 pairs are $sq$ and $pc_4$, and neither singleton–pair incidence is the opposite one.
Suppose the supports are at distance 1: $s$ at $c_0$ is adjacent to $p$ on $\{c_2, c_3\}$, and $t$ at $c_1$ is adjacent to $q$ on $\{c_3, c_4\}$. The path $s, p, c_3, q, t, c_1$ is an induced $P_6$, by the same trace restrictions: the only singleton–pair edges among these vertices are $sp$ and $tq$, the only cycle edges on the path are $pc_3$ and $qc_3$, and $t$ is adjacent to $c_1$ alone on the cycle. Either configuration contradicts the hypothesis that $G$ is $P_6$-free. Therefore $b \leq 1$.
If $b = 0$, then $S = 2n - j - m \leq 2n$. If $b = 1$, then $j \geq 1$ because $b \leq \min(h, j)$, so $S = 2n + 2 - j - m \leq 2n + 1$. In this case $S \leq 2n + 1$.
*Case 3: no holes.* The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong $C_4$-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two,

with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \le 2(n-1) = 2n-2$.
The three cases give the values of $M(n)$. A graph of order $n \ge 11$ cannot realise $k \in \{2,3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs $\Gamma_h$ and the graphs with the same first layer and $m \ge 1$. The latter have $S \le 2n+1$. Cases 2 and 3 give $S \le 2n+1$ and $S \le 2n-2$. Thus $M(n) = 2 + 2/n$, attained only by $\Gamma_{n-7}$.
At order 10 the only graph in the list with $S > 2 \cdot 10 + 2$ is $P$, with $S = 30$. At order 9 the only such graph is $P - v$, with $S = 24$, whether it arises as $(k,h) = (3,0)$ or as $(k,h) = (2,1)$. At order 8 the only such graph is $H_8$, with $S = 20$. At order 7 one has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 16$ is $\Gamma_0$, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \ge 11$, the bound $S \le 3n$ holds for every order $n \ge 2$, with equality only for $P$. The case division does not use $n \ge 7$. At order 4 there is no cycle of length five or six, so only the chordal bound applies and $S \le 6$. Equality holds for a tree, and the only trees on four vertices are $P_4$ and $K_{1,3}$. At order 6 the induced 6-cycle branch contributes only $C_6$, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \le 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph $C_5$ with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly two graphs. □

*Remark 3.7.* Karthick and Maffray describe the structure of graphs with no induced $P_6$ and no induced 4-cycle, and deduce the colouring bound $\chi \le \lceil 5\omega/4 \rceil$ [12]. Their Theorems 3.2 and 3.3 describe the neighbourhood, on an induced $C_5$ or $C_6$, of a vertex outside the cycle. Specialising the 6-cycle description to graphs with no 4-cycle subgraph excludes a consecutive triple, a consecutive quadruple and the full cycle as a support, and leaves an antipodal pair. That is the trace statement at the beginning of Case 1. The constraint on the second layer, the bound $h \le 1$ when $k \in \{2,3\}$, and the exact function $M(n)$ are further arguments. They are not consequences of the local support list, and the colouring theorem is not a substitute for the extremal statement under the subgraph prohibition.

*Remark 3.8.* Brandstädt and Hoàng analysed $(P_6, C_4)$-free graphs through clique cutsets [1]. In that language $C_4$ is an induced cycle. A graph in the class with no clique cutset has every induced $C_5$ dominating, and the presence of an induced $C_6$ that fails to dominate forces the graph to be the join of a blow-up of the Petersen graph and a clique, possibly empty. Theorem 3.6 is a different statement. The prohibition here is that of a 4-cycle as a subgraph, and the conclusion is the exact maximum of $S$ over every connected graph in the resulting class, including graphs that have clique cutsets. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither construction enters the class.

*Remark 3.9.* An alternative proof of $S \le 3n$, with equality only for the Petersen graph, fixes a vertex of eccentricity at most 2 by Lemma 3.5 and charges the cross edges in the second distance layer. The case division above already gives that inequality and the equality case, so the charging argument is not repeated.

## 4 | Radius four

**Lemma 4.1.** *Let $G$ be connected, strong $C_4$-free and $P_8$-free. If an induced 8-cycle dominates $G$ and $r(G) \ge 4$, then $G = C_8$.*

*Proof.* Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex $y$ the trace $A_y = N(y) \cap V(C)$ is nonempty. The trace $A_y$ is not contained in any three consecutive vertices of $C$. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends $c_i$ and $c_{i+2}$ are not both in $A_y$, since $y, c_i, c_{i+1}, c_{i+2}$ would be a 4-cycle. Thus $A_y$ is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact $c_{i+1}$ does not lie on the path. Either path is an induced $P_8$.
Fix a cycle vertex, say $c_0$. The vertices at cycle distance greater than two from $c_0$ are the three consecutive vertices $c_3, c_4, c_5$. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of $c_0$. The same holds with $c_0$ replaced by any vertex of $C$.
Suppose $w$ lies outside $C$, and write $A = N(w) \cap V(C)$. No two vertices of $A$ are at cycle distance two, for they would form a 4-cycle with $w$. A set of vertices of $C$ in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities have been excluded. Therefore $A$ contains two vertices at cycle distance three or four. Rotating, take them to be $c_0$ and $c_d$ with $d \in \{3,4\}$.

The antipode of $c_0$ is $c_4$. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of $C$ is already at cycle distance at most three from $c_0$. Every outside vertex is within distance three of $c_0$. Thus the eccentricity of $c_0$ is at most three, contradicting $r(G) \geq 4$.
Hence $C$ has no vertex outside it, and $G = C_8$. □

*Remark 4.2.* The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced $P_8$. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced $P_8$. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order 8, ... , 13 the counts are 1, 32, 262, 384, 107, 4. It finds 73 isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

**Theorem 4.3.** *Let $G$ be connected and strong $C_4$-free, with $r(G) = 4$ and $p(G) = 7$. Then either $G \cong C_8$, or $G$ is obtained from a 6-cycle $C = v_0 v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each $i$ there is a nonempty set $C_i$ of vertices outside $C$. Every vertex of $C_i$ is adjacent to $v_i$ and to no other vertex of $C$. There is no edge between $C_i$ and $C_j$ for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of $C$ there is at most one further vertex, adjacent precisely to the two ends of that edge.*

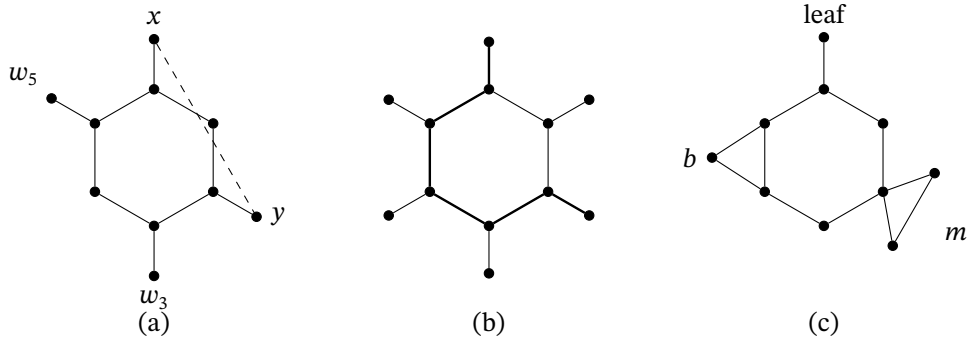



**FIGURE 2** | The radius-four case of Theorem 4.3. (a) If $x$ and $y$ have singleton traces $\{v_0\}$ and $\{v_2\}$ on the dominating 6-cycle, the dashed edge $xy$ completes the induced path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The vertices $w_5$ and $w_3$ are the distance-four witnesses of the proof. (b) One leaf at each cycle vertex gives $S = 2n$. The heavy path has seven vertices. (c) On the same cycle: an isolated leaf, a matching edge inside one set $C_i$, and a vertex glued to a cycle edge. The leaf does not contribute to $b + m$. The matching edge and the glued vertex each increase $2n - S = b + m$ by 1.

*Proof.* Theorem 2.4 gives $p(G) \geq 7$, which agrees with the hypothesis. Apply Theorem 2.5 with $k = 8$. There is a connected dominating set $X$ such that $G[X]$ is $P_6$-free or $G[X] \cong C_8$.
If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.
If $G[X]$ is $P_6$-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in $G$, because $X$ dominates $G$, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.
*Traces.* As in the proof of Lemma 3.4, strong $C_4$-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some $w$ has trace $\{v_i, v_{i+3}\}$, then $v_i$ is within distance two of every vertex of $C$, hence within distance three of every vertex of $G$, a contradiction. So there are no antipodal traces. Let $C_i$ be the set of vertices with trace $\{v_i\}$.
*Witnesses.* Fix $i$. Since $r(G) = 4$, some vertex is at distance at least four from $v_{i+3}$. Every vertex of $C$ is within distance three of $v_{i+3}$, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from $v_{i+3}$, which is at most three unless the trace of $w$ is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from $v_{i+3}$. Every neighbour $z \in W$ of $w_i$ lies in $C_i$: if $z$ had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if $z$ had the neighbour $v_{i-1}$ or $v_{i+1}$, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.
*Edges in $W$.* Figure 2 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced $P_8$.

- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced $P_8$. By symmetry, vertices of $C_i$ and $C_j$ with $i \neq j$ are never adjacent.
- $x$ with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices $v_0, v_1, x, y$ span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle $x, v_0, v_5, y$ (respectively $x, v_1, v_2, y$) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced $P_8$, and $j = 4$ is symmetric.
- $x, y$ with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm 1$ and $j = \pm 2$ give 4-cycles as above; $j = 3$ gives the induced $P_8$ $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside $W$ joins two vertices of the same class $C_i$, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any $C_j$ and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class $C_i$ is nonempty. This is the stated construction.

Conversely, let $G$ be $C_8$, or let $G$ be built from $C = v_0 \cdots v_5$ as in the statement. The cycle $C_8$ has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on $C$. There is no edge between distinct sets $C_i$, and every vertex of $C_i$ meets $C$ only at $v_i$. A cycle that meets $C_i$ is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at $v_i$ together with a matching, so the only cycles are the triangles formed by $v_i$ and one matching edge.

Let $y$ be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through $y$ uses both edges at $y$, and the rest of the cycle is a $v_j$–$v_{j+1}$ path avoiding $y$. A 4-cycle through $y$ is then a path of length two between $v_j$ and $v_{j+1}$ that avoids $y$: the two ends of the edge have a common neighbour other than $y$. No vertex of any $C_i$ is adjacent to two vertices of $C$, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through $y$. The only cycle that uses no vertex outside $C$ is $C$ itself. Thus $G$ has no 4-cycle subgraph.

Let $P$ be an induced path. An isolated vertex of some $G[C_i]$ has degree one, so it is an end of $P$ if it lies on $P$. A vertex of a matching edge in $C_i$ has neighbourhood $\{v_i, x'\}$, a clique. It is therefore an end of $P$ whenever $P$ has a third vertex, and $P$ cannot contain both ends of the matching edge together with $v_i$. A vertex added on a cycle edge has neighbourhood equal to that edge, again a clique, so it too is an end of any induced path that contains it, and the path cannot contain both ends of the cycle edge. Consequently every induced path of order at least three meets $V(G) \setminus V(C)$ in at most two vertices, both ends of the path. The remaining vertices lie on $C$, which has no induced path on six vertices. Hence $p(G) \leq 7$. Since each $C_i$ is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: $x$ and $y$ meet the path only at $v_0$ and $v_2$, and $C$ contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets $C_i$, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along $C$. The distance from $v_i$ to $v_{i+3}$ is therefore three. A vertex of $C_{i+3}$ is at distance four from $v_i$, and the eccentricity of $v_i$ is at least four. It is at most four: every vertex of $C_j$ is at distance $d_C(v_i, v_j) + 1 \leq 4$ from $v_i$, and every vertex added on a cycle edge is within distance four of $v_i$. A vertex of $C_i$ is at distance five from a vertex of $C_{i+3}$. A vertex $y$ added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from $v_j$ and from $v_{j+1}$ to $v_{j+3}$ are 3 and 2, neither neighbour of $y$ is adjacent to $v_{j+3}$, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of $C_{j+3}$ is at distance four from $y$. Every eccentricity is at least four, and $r(G) = 4$. □

**Corollary 4.4.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let $b$ be the number of vertices added on edges of the 6-cycle and let $m$ be the number of matching edges in the sets $C_i$. Then $S(G) = 2n - b - m$. In particular $S(G) \leq 2n$, so $\ell(G) \leq 2$, and* RAD(4) *holds.*

*Proof.* For $G \cong C_8$ one has $n = 8, e = 8, T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, $a$ edges from the sets $C_i$ to the cycle, $m$ matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

**Corollary 4.5.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or $G$ is obtained from $C_6$ by attaching at least one leaf to each cycle vertex.*

*Proof.* By Corollary 4.4, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each $C_i$ is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with $C_8$, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

*Remark 4.6.* If $G$ is connected and strong $C_4$-free with $r(G) = 4$, $p(G) \geq 8$ and $\ell(G) \leq 4$, then $p(G) \geq 8 = r(G) + 4 \geq r(G) + \ell(G)$. This comparison is not a structural description of $G$.

## 5 | Triangle-free graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \dots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; $F_5$ is the Petersen graph, and the parameters $t$ and $t'$ attached to $F_2$ and $F_0'$ are positive integers. The edge counts below use those constructions rather than the accompanying figure.

**Lemma 5.1** (Chen and Xu [5, Lemma 3.4])**.** *Let $G$ be a connected $(P_7, C_4, C_3)$-free graph with no clique cutset. If $G$ is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- $F_1$ is $C_5$. The graph $F_2$ is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets $A_1$ and $A_3$ of equal positive order $t$, with a perfect matching between them, each vertex of $A_1$ adjacent only to $v_1$ and its match, and each vertex of $A_3$ adjacent only to $v_3$ and its match.
- On the same 5-cycle, $F_3$ has three further vertices inducing the path $a_3a_1a_4$, with $a_i$ adjacent to $v_i$. The graph $F_4$ has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph $F_5$ is the Petersen graph.
- $F_0$ is $C_6$. The graph $F_0'$ is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets $Y_1$ and $Y_4$ of equal positive order $t'$, with a perfect matching between them, each vertex of $Y_1$ adjacent only to $v_1$ and its match, and each vertex of $Y_4$ adjacent only to $v_4$ and its match.
- $F_6$ is $C_7$. On a 7-cycle $v_1 \cdots v_7$, write $x_i$ for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then $F_7$ is the cycle plus $x_1$; $F_8$ is the cycle plus nonadjacent $x_1, x_2$; $F_9$ is $F_8$ plus a vertex adjacent only to $x_1$ and $x_2$; $F_{10}$ is the cycle plus nonadjacent $x_1, x_5$; $F_{11}$ is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and $F_{12}$ is $F_{11}$ plus a vertex adjacent only to $x_1$ and $x_2$.

Chen and Xu record that $F_0, F_1$, the graph $F_2$ with $t = 1$, $F_3$ and $F_4$ are induced subgraphs of the Petersen graph, and that $F_6, \dots, F_{11}$ are induced subgraphs of $F_{12}$. The graphs $F_6, \dots, F_{12}$ are the extensions of $C_7$ by vertices of cycle-distance three. The graphs $F_2$ and $F_0'$ are the matching extensions of one pair of opposite vertices on $C_5$ and on $C_6$.

**Lemma 5.2.** *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \leq 3n - 3$. The Petersen graph satisfies $2e = 3n$.*

*Proof.* The constructions give the following orders and sizes. For $F_2$ and $F_0'$ the parameters $t, t'$ are at least one.

| Graph | $n$ | $e$ | $2e$ | $3n - 2e$ |
|---|---|---|---|---|
| $F_1 = C_5$ | 5 | 5 | 10 | 5 |
| $F_2$ | $5 + 2t$ | $5 + 3t$ | $10 + 6t$ | 5 |
| $F_3$ | 8 | 10 | 20 | 4 |
| $F_4$ | 9 | 12 | 24 | 3 |
| $F_5$ | 10 | 15 | 30 | 0 |
| $F_0 = C_6$ | 6 | 6 | 12 | 6 |
| $F_0'$ | $6 + 2t'$ | $6 + 3t'$ | $12 + 6t'$ | 6 |
| $F_6 = C_7$ | 7 | 7 | 14 | 7 |
| $F_7$ | 8 | 9 | 18 | 6 |
| $F_8, F_{10}$ | 9 | 11 | 22 | 5 |
| $F_9, F_{11}$ | 10 | 13 | 26 | 4 |
| $F_{12}$ | 11 | 15 | 30 | 3 |

The five edges of $F_3$ beyond $C_5$ are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of $F_4$ beyond $C_5$ are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex $x_i$ contributes two edges, and each vertex placed on a pair $x_ix_j$ contributes two further edges. The Petersen graph is the only row with difference zero. □

**Lemma 5.3.** *Let $F$ be connected and strong $C_4$-free, and let $s$ be a vertex at which no induced $P_4$ starts. Write $m = e(F[N(s)])$. Then $S(F) \leq 2(|F| - 1) - m$.*

*Proof.* A vertex at distance 3 from $s$ would end a geodesic, hence an induced $P_4$, starting at $s$. So $V = \{s\} \cup N_1 \cup N_2$ with $N_i$ the $i$-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of $y$ with a different parent would give the induced path $sayz$. So $y$ has at most one neighbour in $N_2$, a sibling (two siblings would be common neighbours of $a$ and $y$), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where $C_a$ is the set of children of $a$

and $\bar{a}$ is the neighbour of $a$ in $N_1$, if any. Since $s$ and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F| - 1) - m$. □

**Theorem 5.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if $G$ is the Petersen graph. In particular every connected triangle-free strong $C_4$-free graph of induced-path order at most 6 satisfies $\mathrm{GEN}_3$, and so does every connected bipartite strong $C_4$-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

*Proof.* The proof is by induction on $n = |V(G)|$. The graph $K_2$ has $2e = 2 < 3n$. Suppose $n \ge 3$, and write $S = 2e$, which is legitimate because $G$ is triangle-free.

Suppose first that $v$ is a cut vertex. For each component $C$ of $G - v$ let $\rho_C$ be the largest order of an induced path in $G[C \cup \{v\}]$ that starts at $v$. Paths from two components form an induced path of $G$ on $\rho_C + \rho_{C'} - 1 \le 6$ vertices, so some component $C_1$ has $\rho_{C_1} \le 3$. Let $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with orders $n_1$ and $n_2$. Then $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Lemma 2.3 gives $S(G) = S(G_1) + S(G_2)$. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$. The graph $G_2$ is a smaller connected triangle-free strong $C_4$-free $P_7$-free graph, so the inductive hypothesis gives $S(G_2) \le 3n_2$. Therefore

$$S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

If $G$ has a cut vertex, this estimate gives the theorem, and equality does not hold. Assume for the remainder of the argument that $G$ has no cut vertex.

Suppose that $\{u, v\}$ is a clique cutset of order two. Then $uv$ is an edge. Since neither vertex is a cut vertex, every component of $G - \{u, v\}$ is adjacent to both $u$ and $v$, and there are at least two such components. A shortest $u$–$v$ path through one component has at least four edges: one internal vertex would form a triangle with $uv$, and two internal vertices would form a 4-cycle. In two components, choose shortest paths, and on each path let $x, y, z$ be the three vertices immediately preceding $v$, in that order from the $u$-side. The same choice in the second component produces $x', y', z'$. The path $x, y, z, v, z', y', x'$ is an induced $P_7$. Indeed, each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \le 6$.

Thus $G$ has no clique cutset. A triangle-free graph has no 4-cycle subgraph if and only if it has no induced 4-cycle, so Lemma 5.1 applies: $G$ is complete or belongs to $\mathfrak{F}$. The only complete triangle-free graph on at least two vertices is $K_2$, already strict. Lemma 5.2 gives $2e \le 3n$ for every graph in $\mathfrak{F}$, with equality only for the Petersen graph.

A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong $C_4$-free $P_7$-free graph.

Equality holds only for the Petersen graph. In particular, a graph with a clique cutset lies strictly below the bound, whether the cutset is a single vertex or an edge. □

**Corollary 5.5.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then $G$ contains a triangle.*

*Proof.* If $G$ were triangle-free, Theorem 5.4 would give $S(G) \le 3n$. □

**Lemma 5.6.** *Let $G$ be a graph and let $K$ be a clique cutset of order $t$, with $G - K$ having $k \ge 2$ components. For each component $C$ let $G_C$ be the subgraph induced by $C \cup K$, and write $\Delta(H) = 2|V(H)| - 3 - e(H)$. Then*

$$\Delta(G) = \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].$$

*The bracket equals 1 if $t = 1$ and equals 0 if $t = 2$ or $t = 3$.*

*Proof.* The edges inside $K$ are counted $k$ times among the graphs $G_C$, and $|V(G)| = \sum_C |V(G_C)| - (k-1)t$, so

$$e(G) = \sum_C e(G_C) - (k-1)\binom{t}{2}.$$

Therefore

$$\begin{aligned}\Delta(G) &= 2\sum_C |V(G_C)| - 2(k-1)t - 3 - \sum_C e(G_C) + (k-1)\binom{t}{2} \\ &= \sum_C \big(\Delta(G_C) + 3\big) - 3 - 2(k-1)t + (k-1)\binom{t}{2} \\ &= \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].\end{aligned}$$

The bracket is $0+3-2=1$ for $t=1$, $1+3-4=0$ for $t=2$, and $3+3-6=0$ for $t=3$. □

**Lemma 5.7.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph on $n \geq 2$ vertices, and write $\Delta(G) = 2n-3-e(G)$. Then $\Delta(G) \geq 0$. If $n \geq 4$, then $\Delta(G) \geq 1$, so $e(G) \leq 2n-4$ and $S(G) \leq 4n-8-3T(G)$. In particular $\ell(G) < 4$.*

*Proof.* A copy of $K_4$ contains a 4-cycle, so $\omega(G) \leq 3$. A diamond contains a 4-cycle, so every graph in the class is $(P_7, C_4, \text{diamond})$-free in the sense of Chen, Wu and Xu [4]. The proof is by induction on $n$. For $n=2$ the graph is $K_2$ and $\Delta = 0$. For $n=3$ the maximum of $e$ is 3, attained by $K_3$, and $\Delta \geq 0$.
Let $n=4$. A graph on four vertices with five edges is $K_4$ minus an edge. Labelling the missing edge $cd$, the cycle $a\,c\,b\,d$ uses four edges of the graph. Thus $e \leq 4$ and $\Delta = 5-e \geq 1$.
Now let $n \geq 5$, and assume the two claims for every smaller graph in the class. Every block $G_C$ arising from a clique cutset is itself connected, strong $C_4$-free and $P_7$-free, of order strictly less than $n$.
Suppose first that $v$ is a cut vertex, and that $G-v$ has $k \geq 2$ components. Lemma 5.6 with $t=1$ gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. Each side has order at least 2, so $\Delta(G_i) \geq 0$ by induction, and $\Delta(G) \geq k-1 \geq 1$.
Suppose next that a triangle $K$ is a clique cutset and $G-K$ has $k \geq 2$ components. Each side $G_i$ contains $K$ and at least one further vertex, so its order is at least 4. Lemma 5.6 gives $\Delta(G) = \sum_i \Delta(G_i)$, and induction supplies $\Delta(G_i) \geq 1$. Hence $\Delta(G) \geq k \geq 2$.
Suppose that $\{u, v\}$ is a clique cutset and $G-\{u,v\}$ has $k \geq 2$ components. Lemma 5.6 gives $\Delta(G) = \sum_i \Delta(G_i)$. Induction gives $\Delta(G_i) \geq 0$ for every side, and each side has order at least 3. If every side had deficit 0, then each side would have order 3 and exactly three edges, so each side would be a triangle. Two triangles that share the edge $uv$ span a 4-cycle on their two opposite vertices, which contradicts the hypothesis that $G$ is strong $C_4$-free. Some side therefore has positive deficit, and $\Delta(G) \geq 1$.
Suppose finally that $G$ has no clique cutset. Theorem 1.1 of [4] states that a connected $(P_7, C_4, \text{diamond})$-free graph with no clique cutset is the Petersen graph, or the graph $F$ defined below, or else has minimum degree at most $\max\{2, \omega-1\}$. Here $\omega \leq 3$, so the last quantity is at most 2. The Petersen graph has $n=10$ and $e=15$, hence $\Delta = 2$. The graph $F$ is obtained from a 7-cycle $v_1 \cdots v_7$ by adding an independent set $\{y_1, y_2, y_3\}$ with

$$N(y_1) = \{v_1, v_4, v_5\}, \qquad N(y_2) = \{v_2, v_5, v_6\}, \qquad N(y_3) = \{v_3, v_6, v_7\}.$$

Thus $F$ has $n=10$ and $e=16$, hence $\Delta = 1$. The three triangles of $F$ are $y_1v_4v_5$, $y_2v_5v_6$ and $y_3v_6v_7$. If instead some vertex $v$ has degree at most 2, then $v$ is not a cut vertex, so $G-v$ is connected and lies in the class. Deleting $v$ changes the deficit by

$$\Delta(G) - \Delta(G-v) = 2 - d(v) \geq 0.$$

For $n \geq 5$ the order of $G-v$ is at least 4, so induction gives $\Delta(G-v) \geq 1$ and therefore $\Delta(G) \geq 1$.
Lemma 2.1(3) gives $S = 2e-3T$. For $n \geq 4$ one has $e \leq 2n-4$, so $S \leq 4n-8-3T \leq 4n-8$ and $\ell \leq 4 - 8/n < 4$. For $2 \leq n \leq 3$ one has $e \leq 2n-3$ and $S \leq 4n-6$, so again $\ell < 4$. □

**Lemma 5.8.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph.*

1. *If $n \geq 5$ and $T(G) \leq 1$, then $\Delta(G) \geq 2$, so $e(G) \leq 2n-5$. If $T(G) = 1$, then $S(G) \leq 4n-13$.*
2. *If $n \geq 4$ and $T(G) \geq 2$, then $S(G) \leq 4n-14$.*
3. *If $n \geq 5$ and $T(G) \geq 1$, then $S(G) \leq 4n-13$.*

*Proof.* If $T \geq 2$ and $n \geq 4$, Lemma 5.7 gives $e \leq 2n-4$, hence $S = 2e-3T \leq 2(2n-4)-6 = 4n-14$. This is (2). Once (1) is known, $T=1$ gives $S = 2e-3 \leq 4n-13$, and (2) is at least as strong, so (3) follows. It remains to prove $\Delta \geq 2$ when $n \geq 5$ and $T \leq 1$.
The proof is by induction on $n$. Let $n=5$. A chordless cycle in a strong $C_4$-free graph has length at least 5, so the only possible hole on five vertices is a 5-cycle. A chord of $C_5$ produces a 4-cycle, and there is no room for a further vertex, so $G \cong C_5$. Then $e=5$, $T=0$ and $\Delta = 2$. If $G$ has no hole, then $G$ is chordal. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and its neighbourhood is a clique. That clique has order at most two: a simplicial vertex of degree three or more, together with three neighbours, contains a $K_4$ and therefore a 4-cycle. Deleting a simplicial vertex of degree one removes one edge and no triangle. Deleting a simplicial vertex of degree two removes two edges and exactly one triangle. Repeating until a single vertex remains gives $e = n-1+T$. Here $T \leq 1$, so $e \leq 5$ and $\Delta = 7-e \geq 2$.
Now let $n \geq 6$, with $T(G) \leq 1$, and assume the deficit claim for every smaller graph in the class that has at most one triangle. By Lemma 5.7 every graph in the class of order at least 2 has nonnegative deficit, and every graph of order at least 4 has deficit at least 1.

Suppose $v$ is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma 5.6 gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. If this quantity were at most 1, then $k = 2$ and both sides would have deficit 0. A side of deficit 0 has order at most 3, so $n \leq 3+3-1 = 5$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose a triangle is a clique cutset, with $k \geq 2$ sides. Each side has order at least 4, so each has deficit at least 1 by Lemma 5.7, and Lemma 5.6 gives $\Delta(G) \geq 2$.
Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components, and suppose toward a contradiction that $\Delta(G) \leq 1$. Lemma 5.6 gives $\Delta(G) = \sum_i \Delta(G_i)$. A side of order at least 4 has deficit at least 1. A side of deficit 0 therefore has order 3 and three edges, so it is a triangle. Two such sides would give two triangles and a 4-cycle, which is impossible under $T(G) \leq 1$ and the strong $C_4$-free hypothesis. If every side had deficit at least 1, the sum would be at least $k \geq 2$. Hence exactly one side is a triangle, of deficit 0, and exactly one other side is present: a third side would contribute deficit at least 1 and would push the sum to at least 2. The hypothesis $T(G) \leq 1$ forces the other side $G_2$ to be triangle-free. Its order is $n - 1 \geq 5$. The inductive hypothesis gives $\Delta(G_2) \geq 2$, so $\Delta(G) \geq 2$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose finally that $G$ has no clique cutset. The Petersen graph has $T = 0$ and $\Delta = 2$. The exceptional graph $F$ of the previous proof has three triangles, so it does not fall under $T \leq 1$. In every remaining case Chen, Wu and Xu supply a vertex $v$ of degree at most 2. This vertex is not a cut vertex. It lies in at most one triangle, and $T(G - v) = T(G) - t(v) \leq 1$. The graph $G - v$ is connected, strong $C_4$-free and $P_7$-free, of order at least 5. Induction gives $\Delta(G - v) \geq 2$. Since $\Delta(G) = \Delta(G - v) + 2 - d(v) \geq \Delta(G - v)$, one has $\Delta(G) \geq 2$. □

**Corollary 5.9.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then $T(G) \geq 1$, the order $n$ is at least 14, and*

$$3 < \ell(G) \leq 4 - \frac{13}{n}.$$

*In particular $\ell(G) \neq 4$, and no graph of order at most 13 lies in this cell.*

*Proof.* Corollary 5.5 supplies a triangle, so $T(G) \geq 1$. Since $p(G) = 6$, the graph is $P_7$-free. If $n \leq 3$, then $S \leq 4$ and $\ell \leq 4/3$. If $n = 4$, Lemma 5.7 gives $e \leq 4$. A triangle-free graph on four vertices in the class is a tree, so $S \leq 6$ when $T = 0$, while $T \geq 1$ gives $S \leq 5$. In all of these cases $\ell \leq 2$, contradicting $\ell > 3$. Thus $n \geq 5$. Lemma 5.8 gives $S \leq 4n - 13$, so $\ell \leq 4 - 13/n$. The inequality $S > 3n$ then forces $n \geq 14$. For every such order one has $4 - 13/n < 4$. □

*Remark 5.10.* The degree bound in Theorem 1.1 of Chen, Wu and Xu does not by itself yield $S \leq 3n$. Lemmas 5.7 and 5.8 are the deficit estimates that follow from that theorem once clique cutsets of order 1, 2 and 3 are restored and the two exceptional graphs are counted. The resulting bound is $\ell < 4$. If at least one triangle is present and $n \geq 5$, it improves to $S \leq 4n - 13$. It is not the bound $S \leq 3n$. The case of $\mathrm{GEN}_3$ in which triangles are present remains open. Appendix A records a conditional reduction for that case.

**Theorem 5.11** (Theorem A)**.** *Let $G$ be connected and strong $C_4$-free with $n \geq 2$. Then $p(G) \geq r(G) + \ell(G)$ whenever $\ell(G) \leq 3$, whenever $r(G) \leq 2$ and $\ell(G) \leq 4$, whenever $r(G) = 4$ and $\ell(G) \leq 4$, and whenever $r(G) \geq 5$ and $\ell(G) \leq 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \leq 2$.*

*Proof.* Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Corollary 4.4. For the inequality we need $p \geq r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \leq 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \leq 4$, then Lemma 3.3 gives $\ell \leq 2$, a contradiction. So $p \geq 5$.
- $r = 2$, $c = 4$: if $p \leq 5$, then Theorem 3.6 gives $\ell \leq 3$, a contradiction. So $p \geq 6$.
- $r = 3$, $c \leq 2$: Theorem 2.4 gives $p \geq 5$.
- $r = 3$, $c = 3$: if $p \leq 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \geq 6$. Theorem 3.6 gives $\ell \leq 3$ for every $P_6$-free graph in the class, which does not rule out $c = 3$.
- $r \geq 4$, $c \leq 3$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 3$.
- $r = 4$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 7$. If $p = 7$, Corollary 4.4 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$, Remark 4.6 gives the same inequality.
- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + c$.

□

Equality in REAL-133 occurs for $C_5$ ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and $C_6$ ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

## 6 | The reduction and the residual cases

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

$\mathrm{GEN}_K$: every connected strong $C_4$-free graph $G$ with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.

$\mathrm{RAD}(r)$: every connected strong $C_4$-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Corollary 4.4 is the case $r = 4$ of $\mathrm{RAD}(r)$, in the stronger form $\ell \leq 2$.

**Corollary 6.1.** *Let $G$ be a connected strong $C_4$-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \geq 3$. Then either*

*(a) $r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so $G$ violates $\mathrm{GEN}_K$; or*
*(b) $r(G) = K + 1 \geq 4$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so $G$ violates $\mathrm{RAD}(K + 1)$.*

*Proof.* Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 3$ by Theorem 5.11. A violation means $p \leq r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \leq p \leq r + K$, so $r \leq K + 1$. If $r \leq K$, then $p \leq r + K \leq 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \leq p \leq 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. □

Since $\mathrm{GEN}_K$ ignores the radius, it is stronger than alternative (a). The statement $\mathrm{GEN}_K$ might fail for large $K$ even if REAL-133 holds; see Remark 8.5.

**Proposition 6.2.** *For $k \geq 3$ let $\theta_k$ consist of two vertices $x, y$ joined by $k$ internally disjoint paths $xa_ib_iy$. Then $\theta_k$ is strong $C_4$-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

*Proof.* Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with $j, i, l$ distinct) is induced. Conversely, let $P$ be an induced path. If $P$ avoids $x$ and $y$, it lies inside one edge $a_ib_i$. If $P$ contains $x$ but not $y$, then $x$ has at most two neighbours on $P$, each followed on $P$ by at most its $b$-vertex, so $|P| \leq 5$. If $P$ contains $x$ and $y$, then every $a_j$ on $P$ is adjacent on $P$ to $x$ and every $b_j$ to $y$, at most one branch lies entirely on $P$, and so $P$ consists of $x$, $y$, at most one full branch and at most one further neighbour of each of $x$ and $y$. Hence $|P| \leq 6$. □

Thus no argument with a uniform positive margin in $\ell$ can prove $\mathrm{GEN}_3$, already among graphs of radius 3. The resulting division of REAL-133 is recorded once, in Table 1.
Let $3 < \ell \leq 4$, and suppose $G$ violates REAL-133. Corollary 6.1 gives $K = 3$, so either $r \leq 3$ and $p \leq 6$, or $r = 4$ and $p = 7$ with $\ell > 3$. The second alternative contradicts Corollary 4.4. If $r \leq 2$, Theorem 5.11 applies. Thus $r = 3$ and $p \leq 6$. Theorem 2.4 gives $p \geq 5$. If $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$. Hence $p = 6$. If instead $p \geq 7$, then $p \geq 7 \geq r + \ell$. For $r \geq 5$ and $\ell \leq 4$, Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4$, so REAL-133 holds with no further structural work. Every counterexample in the range $\ell \leq 4$ therefore satisfies $r = 3$ and $p = 6$. Corollary 5.9 sharpens the cell to order at least 14, with $T \geq 1$ and $3 < \ell \leq 4 - 13/n$. In particular $\ell = 4$ does not occur. The parts of the argument that are not the Erdős–Saks–Sós bound are Theorems 3.6, 4.3 and 5.4, together with Lemmas 5.7 and 5.8. The inequality $\mathrm{GEN}_3$ for graphs that contain a triangle would close the remaining cell. It is not proved in this paper; see Appendix A.

**Lemma 6.3.** *Let $G$ be connected, bipartite and strong $C_4$-free, with $r(G) = 2$. Then $G$ is a tree, $\ell(G) = 2 - 2/n$ and $p(G) \leq 5$.*

*Proof.* Let $s$ be a centre, and let $A$ be the colour class of $s$ and $B$ the other class. Every vertex of $B$ lies at odd distance from $s$, hence at distance 1, so $B \subseteq N(s)$. Every vertex $x \in A \setminus \{s\}$ lies at distance 2 from $s$, so $x$ has a neighbour in $B$. Two such neighbours would form a 4-cycle with $s$. Thus $x$ has exactly one neighbour. There is no edge inside $A$ or inside $B$, and the number of edges is $|B| + |A| - 1 = n - 1$. A connected graph with $n - 1$ edges is a tree. There are no triangles, so $S = 2e = 2n - 2$ and $\ell = 2 - 2/n$. Every vertex of $A \setminus \{s\}$ is a leaf. An induced path contains at most two leaves, and the unique path between them has the form leaf, neighbour in $B$, $s$, neighbour in $B$, leaf. Hence $p(G) \leq 5$. □

**Lemma 6.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_8$-free graph. Then $e(G) \leq 2n - 2$. In particular, for $n \geq 2$ one has $\ell(G) = 2e(G)/n < 4$.*

*Proof.* The graph has no triangle and no 4-cycle subgraph, so every cycle has length at least 5. The bound $e \leq 2n - 2$ holds for $n = 1$. Let $n \geq 2$, and proceed by induction on the order. If $G$ is a tree, then $e = n - 1 \leq 2n - 2$. Otherwise the girth is at least 5. Theorem 1 of Chudnovsky and Stacho [6] then asserts that $G$ is the Petersen graph, the Heawood graph, the graph

obtained by contracting one edge of the Heawood graph, or else $G$ has a vertex of degree at most 2. These three exceptions have pairs $(n, e)$ equal to $(10, 15)$, $(14, 21)$ and $(13, 20)$. Each satisfies $e \le 2n - 2$, since $15 \le 18$, $21 \le 26$ and $20 \le 24$.
In every remaining case some vertex $v$ has degree at most 2. Every component $H$ of $G - v$ is connected, triangle-free, strong $C_4$-free and $P_8$-free. The inductive hypothesis gives $e(H) \le 2|H| - 2$. If $G - v$ has $c \ge 1$ components, then

$$e(G) = e(G - v) + d(v) \le 2(n - 1) - 2c + 2 \le 2n - 2,$$

because $d(v) \le 2$ and $c \ge 1$. Since $G$ is triangle-free, $S = 2e \le 4n - 4$, so $\ell \le 4 - 4/n < 4$. □

Lemma 6.3 excludes the pair $(r, p) = (2, 7)$ for every connected bipartite strong $C_4$-free graph: the induced-path order is at most 5, and $\ell < 2$. Lemma 6.4 is the same exclusion for every connected triangle-free graph in the class, and it also excludes $(r, p) = (3, 7)$ from the range $5 < \ell \le 6$. A graph with $p = 7$ is $P_8$-free, so a triangle-free graph of that path order has $\ell < 4$. Thus any graph with $(r, p) \in \{(2, 7), (3, 7)\}$ and $5 < \ell \le 6$ contains a triangle. Lemma 5.7 excludes $(2, 6)$ and $(3, 6)$ for every connected strong $C_4$-free graph: $p \le 6$ forces $\ell < 4$.

## 7 | Rooted bounds of orders three and four

The arguments of this section and of Sections 8 and 9 are not used for Theorems 3.6, 4.3 and 5.4 or for Corollary 5.9. The rooted bound that bears on the next open band is Theorem 7.2: it is the estimate which shows that a minimum counterexample to $\mathrm{GEN}_4$ is 2-connected. The distance-layer form of the same estimate is omitted. The block-class bound and the Moore observation are kept because they delimit $\mathrm{GEN}_K$, and Example 9.4 is kept because it is the explicit failure of the endpoint inequalities at order nine.
For a vertex $s$ of a connected graph $F$ let $p_s(F)$ be the largest order of an induced path starting at $s$. Write $D = d_F(s)$ and $m = e(F[N(s)])$.

**Definition 7.1.** For an integer $q \ge 3$, the rooted bound $\mathrm{ROOT}(q)$ asserts that every connected strong $C_4$-free graph $F$ and every vertex $s$ with $p_s(F) \le q$ satisfy

$$S(F) \le (q - 1)(|F| - 1) - (q - 3)D - m.$$

The hypothesis $p_s(F) \le q$ belongs in the statement. Without it the displayed inequality is false: if $q = 3$ and $F = C_5$, then $S(F) = 10$ and $m = 0$, while $2(|F| - 1) = 8$. Lemma 5.3, already proved because the cut-vertex induction of Theorem 5.4 uses it, is the case $q = 3$.

**Theorem 7.2.** *Let $F$ be connected and strong $C_4$-free, and let $s \in V(F)$ satisfy $p_s(F) \le 4$. Write $D = d(s)$ and $m = e(F[N(s)])$. Then*

$$S(F) \le 3(n - 1) - D - m. \tag{7.1}$$

*In particular* ROOT(4) *holds. Equality holds for $C_5$ at any root, and for a graph formed by $k \ge 1$ copies of $C_5$ that meet only at the root.*

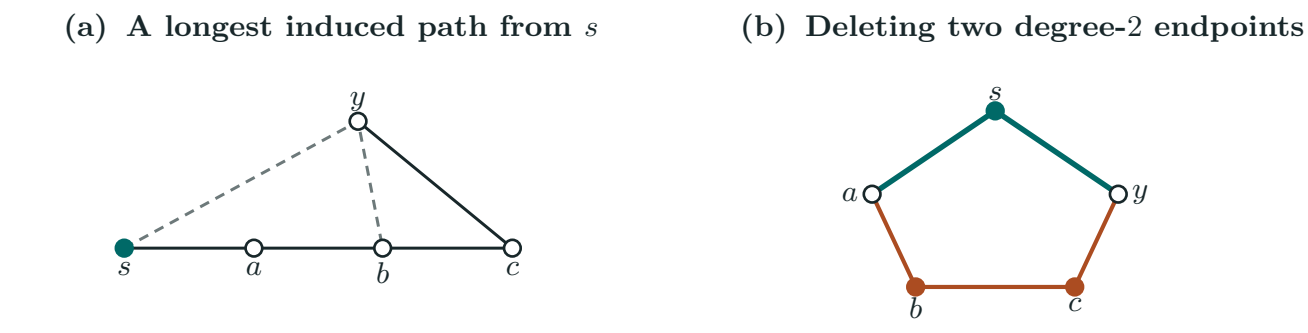



**FIGURE 3** | Deletion at the end of a longest induced path of order four starting at $s$. (a) Each neighbor $y \ne b$ of $c$ must be adjacent to $s$ or $b$. Dashed lines indicate alternatives. (b) If neither terminal vertex $c$ nor $b$ can be deleted with loss at most three, the indicated 5-cycle is induced and both have degree two. The deletion of $\{b, c\}$ has loss six and preserves connectivity.

*Proof.* The proof is by induction on $n$. If $p_s(F) \le 3$, Lemma 5.3 gives $S \le 2(n-1) - m$. Since $D \le n-1$, the right-hand side is at most $3(n-1) - D - m$.
Now suppose $p_s(F) = 4$, and let $s, a, b, c$ be an induced path. The endpoint $c$ is not a cut vertex: a neighbour of $c$ in a component of $F - c$ that does not contain $s$ would extend the path. Let $y \in N(c) \setminus \{b\}$. The path $s, a, b, c, y$ is not induced, so $y$ meets at least one of $s, a, b$. The edge $ya$ would give the 4-cycle $a, b, c, y$. Thus $y$ is adjacent to $s$ or to $b$. The vertices $s$ and $c$ have at most one common neighbour, and so do $b$ and $c$, hence $d(c) \le 3$. If $d(c) = 3$, then $c$ has two neighbours besides $b$, and one of them is adjacent to $b$. So $c$ lies in a triangle, $t(c) \ge 1$ and $L(c) = 2d(c) - 3t(c) \le 3$. Figure 3 shows the two deletions.
If $L(c) \le 3$, delete $c$. The graph $F - c$ is connected, and $c \notin N[s]$, so $D$ and $m$ are unchanged. The inductive hypothesis gives $S(F - c) \le 3(n-2) - D - m$, and therefore $S(F) \le 3(n-2) - D - m + 3 = 3(n-1) - D - m$.
The remaining case is $d(c) = 2$ and $t(c) = 0$, so $L(c) = 4$. Let $y$ be the neighbour of $c$ other than $b$. Then $y$ is not adjacent to $b$, hence $ys$ is an edge, and $s, a, b, c, y$ is an induced 5-cycle: every chord is excluded by the induced path $s, a, b, c$, by $t(c) = 0$, or by the 4-cycle that $ay$ would create. The vertex $b$ is the endpoint of the induced path $s, y, c, b$. If $L(b) \le 3$, delete $b$ and apply the inductive hypothesis as above.
Otherwise the same endpoint analysis gives $d(b) = 2$ and $t(b) = 0$. Delete $U = \{b, c\}$. A path that used the edges $ab, bc, cy$ can be replaced along $a, s, y$, so $F - U$ is connected. The deleted edges are $ab$, $bc$ and $cy$, and no triangle is deleted. Since $S = 2e - 3T$, one has $D_F(U) = 6 = 3|U|$. The vertices $b$ and $c$ lie outside $N[s]$, so $D$ and $m$ are unchanged. Induction gives $S(F - U) \le 3(n-3) - D - m$, and $S(F) \le 3(n-3) - D - m + 6 = 3(n-1) - D - m$.
For the equality examples, root $C_5$ at any vertex. Then $n = 5$, $D = 2$, $m = 0$ and $S = 10 = 3(n-1) - D - m$. If $k$ copies of $C_5$ meet only at the root, then $n = 1 + 4k$, $D = 2k$ and $m = 0$, because the two neighbours of the root on each copy are nonadjacent, and $S = 10k = 3(n-1) - D - m$. □

**Proposition 7.3.** *Let $F$ be connected and strong $C_4$-free, and let $s$ satisfy $p_s(F) = 4$ and $\mathrm{ecc}(s) = 3$. Then $S(F) \le 3(n-1) - D - m - 2$. Equality holds for $P_4$ rooted at an endpoint.*

*Proof.* The proof is by induction on $n$ for instances with $p_s = 4$ and $\mathrm{ecc}(s) = 3$. Fix a vertex $z$ at distance 3 from $s$. Whenever an endpoint other than $z$ exists, the deletion in the proof of Theorem 7.2 applies to it. In the exceptional 5-cycle the vertices $b$ and $c$ are at distance 2 from $s$, so the deletion leaves $z$. The remaining graph is connected and strong $C_4$-free, $D$ and $m$ are unchanged, and distances do not decrease. Thus $d(s, z) \ge 3$. Since $p_s \le 4$, no distance exceeds 3, so $p_s = 4$ and the eccentricity of $s$ is still 3. A deletion of one vertex of loss at most 3 contributes 3, and the inductive bound on $n-1$ vertices contributes $3(n-2) - D - m - 2$, which sum to $3(n-1) - D - m - 2$. Deleting $\{b, c\}$ contributes 6 against $3(n-3) - D - m - 2$.
If $z$ is the unique endpoint, delete $z$. Then $p_s(F - z) \le 3$. An extra neighbour of $z$ is not adjacent to $s$, for that would give $d(s, z) \le 2$. By the endpoint analysis it is adjacent to the predecessor of $z$ and forms a triangle, so $L(z) \le 2$. Lemma 5.3 gives $S(F - z) \le 2(n-2) - m$, hence $S(F) \le 2(n-1) - m$. A geodesic from $s$ to $z$ has at least two vertices outside $N[s]$, so $n - 1 - D \ge 2$ and $2(n-1) - m \le 3(n-1) - D - m - 2$.
For $P_4$ rooted at an endpoint one has $n = 4$, $D = 1$, $m = 0$ and $S = 6$, and the strengthened bound is $3 \cdot 3 - 1 - 2 = 6$. □

**Proposition 7.4.** *Let $G$ be a counterexample to $\mathrm{GEN}_4$ of minimum order. Then $G$ is 2-connected. This does not decide $\mathrm{GEN}_4$.*

*Proof.* Such a graph is connected and strong $C_4$-free, with $p(G) \le 8$ and $S(G) \ge 4n + 1$. Let $v$ be a cut vertex, and let $G_1, G_2$ be the graphs of Lemma 2.3. Write $\rho_i$ for the largest order of an induced path of $G_i$ that starts at $v$. Paths from the two sides form an induced path of $G$ on $\rho_1 + \rho_2 - 1 \le 8$ vertices, so some side, say $G_1$, has $\rho_1 \le 4$. Theorem 7.2 gives $S(G_1) \le 3(n_1 - 1)$. The graph $G_2$ is a smaller connected strong $C_4$-free graph of path order at most 8, so minimality gives $S(G_2) \le 4n_2$. Therefore $S(G) \le 3(n_1 - 1) + 4n_2$. Since $n = n_1 + n_2 - 1$,

$$3(n_1 - 1) + 4n_2 = 4n - n_1 + 1.$$

The hypothesis $n_1 \ge 2$ gives $4n - n_1 + 1 \le 4n - 1$, so $S(G) < 4n$, contradicting $S(G) \ge 4n + 1$. □

For $q \ge 5$ the bound $\mathrm{ROOT}(q)$ is not proved for arbitrary strong $C_4$-free graphs. Section 10 records the limitation. The next section proves every index for graphs whose blocks are bipartite or cycles. Conjecture 10.2 would imply $\mathrm{ROOT}(q)$ for the same $q$, by the attachment in Section 10. That implication is not a proof of the conjecture, and for $q \ge 5$ it does not prove $\mathrm{ROOT}(q)$.

## 8 | Rooted bounds for every index

Let $\mathcal{B}$ be the class of connected strong $C_4$-free graphs in which every block is bipartite or a cycle. Edges are blocks, and cycles may be odd. The class contains every connected strong $C_4$-free bipartite graph and every connected strong $C_4$-free

cactus. Theorem 8.2 keeps this block hypothesis. The graph $K_{2,3}$ does not show that the hypothesis is necessary. Its only block is itself, and that block is bipartite, but $K_{2,3}$ contains a 4-cycle subgraph. Rooted in its part of size two it has $p_s = 3$ and $S = 12 > 2(n-1) - m = 8$. The example tests the failure of ROOT(3) after the strong $C_4$-free hypothesis is removed. It does not test the block restriction inside the strong $C_4$-free class.

A rooted graph $(G, s)$ with $p_s(G) = q$ and $\mathrm{ecc}(s) = q-1$ is called *terminal*. For a terminal instance put $R = n-q+1-D \geq 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor (q-1)/2 \rfloor$ for $q \geq 5$.

**Lemma 8.1.** *Let $G$ be connected and strong $C_4$-free, and let $P$ be a longest induced path starting at a vertex $s$, of order $q$. Let $x$ be the other end of $P$, and suppose that the block $B$ containing $x$ is bipartite. Let $Q$ be the subpath of $P$ formed by the vertices of $P$ that lie in $B$, and set $\ell = |Q|$. Then every neighbour of $x$ lies in $B$, and $x$ lies in no triangle. If $\ell = 2$, then $d(x) = 1$. If $\ell \geq 3$, then $d(x) \leq \lfloor (\ell-1)/2 \rfloor$. In every case $\ell \leq q$, and $L(x) \leq c_q$ whenever $q \geq 5$.*

*Proof.* A neighbour of $x$ in a component of $G - x$ that does not contain $s$ would extend $P$, so $x$ is not a cut vertex. Every edge with one end in $B$ and one end outside $B$ is incident with a cut vertex of $B$. Therefore every neighbour of $x$ lies in $B$. The block is bipartite, so $x$ lies in no triangle.

Since the block-cut tree is a tree, $P$ enters $B$ only once. The intersection of $P$ with $B$ is a single subpath $Q = z_0, \dots, z_{\ell-1}$ with $z_{\ell-1} = x$. If $s$ lies in $B$, then $Q = P$. If $s$ lies outside $B$, the vertices of $P$ that precede $z_0$ lie outside $B$, and $z_0$ is their only neighbour in $B$: any other edge from that prefix into $B$ would be a second entry. A vertex of $B \setminus \{z_0\}$ is therefore adjacent to no vertex of the prefix. An extension of $P$ at $x$ through a vertex of $B$ can fail only by a chord into $Q$.

Let $y$ be a neighbour of $x$ other than $z_{\ell-2}$. Then $y$ lies in $B$, and $y$ does not lie on $Q$, because $Q$ is induced. The vertex $y$ is not adjacent to the prefix. If $y$ had no neighbour on $Q$, the path $P$ followed by $y$ would be induced. Thus $y$ has a neighbour $z_i$ on $Q$. This neighbour is not $x$. It is not $z_{\ell-2}$, for $yxz_{\ell-2}$ would be a triangle, and it is not $z_{\ell-3}$, for $z_{\ell-3}z_{\ell-2}xy$ would be a 4-cycle. Since $B$ is bipartite, $z_i$ has the colour of $x$, so $i \equiv \ell - 1 \pmod 2$ and $i \leq \ell - 5$.

If $\ell = 2$, then $Q$ is a single edge and there is no admissible index, so $d(x) = 1$. If $\ell \geq 3$, the number of admissible indices is $\lfloor (\ell-1)/2 \rfloor - 1$. Two neighbours of $x$ are nonadjacent and cannot share a blocker, for a common blocker would form a 4-cycle with them and with $x$. Hence $d(x) \leq \lfloor (\ell-1)/2 \rfloor$. In every case $\ell \leq q$. For $q \geq 5$ one has $d(x) \leq \lfloor (q-1)/2 \rfloor$, and $L(x) = 2d(x) \leq c_q$.

The same count applies inside $B$. Let $z$ be the cut vertex of $B$ lying on every path from a fixed root to $B$, or the root itself if the root lies in $B$, and let $Q$ be a longest induced path of $B$ starting at $z$, with other end $x$ and with $\ell = |Q|$. A neighbour of $x$ in $B$ that had no neighbour on $Q$ would extend $Q$ inside $B$. If the root lies outside $B$, no vertex of $B \setminus \{z\}$ meets the component of the root at $z$, so that component contributes no further blocker. Thus $d_B(x) \leq 1$ if $\ell = 2$, and $d_B(x) \leq \lfloor (\ell-1)/2 \rfloor$ if $\ell \geq 3$. □

**Theorem 8.2.** *1. Every terminal instance in $\mathcal{B}$ satisfies $S(G) \leq 2(D + q - 2) - m + c_q R$.*

*2. Every $F \in \mathcal{B}$ with a vertex $s$ such that $p_s(F) \leq q$ satisfies*

$$S(F) \leq c_q(n-1) - (c_q - 2)D - m.$$

*In particular* ROOT($q$) *holds in $\mathcal{B}$ for every $q \geq 3$. If every block is bipartite or a triangle, one may take $c_q = 2\lfloor (q-1)/2 \rfloor$ for all $q \geq 3$.*

*Proof. Two terminal facts.* An endpoint $x$ of a longest rooted induced path is not a cut vertex: a neighbour of $x$ in another component of $G - x$ would extend the path. If $z$ is at distance $q-1$ from $s$ and $v_0 = s, \dots, v_{q-1} = z$ is a geodesic, any neighbour $y \neq v_{q-2}$ of $z$ must meet an earlier path vertex (else the rooted path extends); meeting $v_i$ with $i \leq q-4$ shortens $d(s, z)$, and meeting $v_{q-3}$ gives a $C_4$; so $y \sim v_{q-2}$, and two such $y$ would give a $C_4$. Hence $L(z) = 2d(z) - 3t(z) \leq 2$. Moreover, if $x \neq s, z$, $G - x$ is connected and $z$ survives, then $q - 1 \leq d_{G-x}(s, z) \leq p_s(G - x) - 1 \leq q - 1$, so the instance stays terminal with the same $q$.

*Endpoint losses.* Let $x$ be the endpoint of a longest rooted path of order $q$. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, and if a cycle of length at least 5 it is 4. If the block is bipartite, Lemma 8.1 gives $L(x) \leq c_q$ for $q \geq 5$.

*No nontrivial bipartite block when $q \leq 4$.* Let $B$ be a nontrivial bipartite block, and let $z$ be the cut vertex of $B$ that lies on every path from $s$ to $B$, or let $z = s$ if $s \in B$. Let $Q$ be a longest induced path of $B$ starting at $z$, and let $x$ be its other end. Every vertex of a nontrivial block has degree at least two in the block, so $d_B(x) \geq 2$. The last paragraph of Lemma 8.1 gives $d_B(x) \leq 1$ if $|Q| \leq 4$ and $d_B(x) \leq \lfloor (|Q|-1)/2 \rfloor$ in any case with $|Q| \geq 3$. The inequality $d_B(x) \geq 2$ therefore forces $|Q| \geq 5$. An induced $s$–$z$ path meets $B$ only at $z$. Concatenating it with $Q$ yields an induced path that starts at $s$. The order of this path is $|Q|$ if $s \in B$, and at least $|Q| + 1$ if $s \notin B$. In either case $p_s(G) \geq 5$. A terminal instance with $q \leq 4$ therefore has no nontrivial bipartite block. Every block is an edge, a triangle, or a cycle of length at least five, so the graph is a cactus.

*Bases.* For $q \le 4$ the preceding paragraph leaves only cacti. For a $C_4$-free cactus with $c_3$ triangle blocks and $c_L$ longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \le c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a $C_5$ through $s$. A chosen length-3 geodesic uses at most one edge of each such $C_5$: if it used two edges and then left the cycle, replacing that two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each $C_5$ contributes two vertices to $R$, $R \ge 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \ge 0$. For $q = 2$ the root is universal and $S = 2D - m$.
*Induction on n.* Let $E$ be the set of endpoints of rooted induced paths of order $q$ and $z$ a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, $x$ is not a cut vertex, its loss is at most $c_q$, the instance remains terminal with the same $q$, $D$ and $m$, and $R$ drops by one. If $E = \{z\}$, delete $z$: the new rooted maximum is $q-1$, the eccentricity $q-2$, $R$ is unchanged and the loss is at most 2; induction gives $S \le 2(D+q-3) - m + c_{q-1}R + 2 \le 2(D+q-2) - m + c_qR$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.
(2) Attach a new path with $q-1$ edges at $s$. The result is terminal of index $q$, lies in $\mathcal{B}$, has order $n+q-1$, mass $S+2(q-1)$ and root degree $D+1$, and $m$ is unchanged. Substituting into (1) gives (2). Since $c_q \le q-1$ and $n-1-D \ge 0$,

$$\begin{aligned} c_q(n-1) - (c_q-2)D - m &= c_q(n-1-D) + 2D - m \\ &\le (q-1)(n-1-D) + 2D - m \\ &= (q-1)(n-1) - (q-3)D - m, \end{aligned}$$

so ROOT($q$) follows. □

A matching upper bound of shape $S \le (r-1)n - 2r(r-3)$, for graphs in $\mathcal{B}$ of radius $r \ge 5$ and path order $2r-1$, has been proposed from a structural description of radius-extremal graphs. That description is not proved in this paper, and the radius bound in $\mathcal{B}$ is not claimed.

**Lemma 8.3.** *Let $G$ be strong $C_4$-free and triangle-free, and let $x_0x_1 \cdots x_k$ be an induced path that cannot be extended at $x_k$. If $d(x_k) \ge 2$, then $k \ge d(x_k) + 1$. If moreover $G$ is bipartite, then $k \ge 2d(x_k)$.*

*Proof.* Let $y \ne x_{k-1}$ be a neighbour of $x_k$. It is not on the path, and, since the path cannot be extended, $y$ has a neighbour $x_i$ with $i \le k-1$. Here $i \ne k-1$ (a triangle) and $i \ne k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_ky$), so $i \le k-3$. A vertex $x_i$ is adjacent to at most one neighbour of $x_k$. Hence the $d(x_k) - 1 \ge 1$ such neighbours need distinct blockers among $x_0, \dots, x_{k-3}$, and $k - 2 \ge d(x_k) - 1$. If $G$ is bipartite, the blocker has the colour of $x_k$, so $i \equiv k \pmod 2$ and $i \le k-4$; there are $\lfloor k/2 \rfloor - 1$ such positions, and $\lfloor k/2 \rfloor \ge d(x_k)$. □

**Proposition 8.4.** *Every connected regular bipartite strong $C_4$-free graph satisfies* $\mathrm{GEN}_K$ *for every $K$ and satisfies REAL-133. Every Moore graph of diameter 2 and girth 5 satisfies REAL-133.*

*Proof.* Let $G$ be $d$-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \ge 2$. As $G$ is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma 8.3 applied to a longest induced path gives $p \ge 2d + 1$, and Theorem 2.4 gives $p \ge 2r - 1$; so $p \ge r + d$ both when $r \le d + 1$ and when $r \ge d + 1$. If $p \le 2K$, then $2d + 1 \le 2K$, so $S = dn < Kn$. A Moore graph of diameter 2 and girth 5 has degree $d \ge 2$, radius 2 and $\ell = d$. The degree is an integer, so the floor form and the real form ask for the same inequality. The first part of Lemma 8.3 gives $p \ge d + 2 = r + \ell$. This is an alternate proof of the case already recorded by Chen [3] immediately after his Theorem 3.6. It is not a new extremal classification. □

*Remark 8.5.* A Moore graph of degree 57, if it exists, has $n = 57^2 + 1 = 3250$ and $S = 57n$. Proposition 8.4 gives REAL-133, which only needs $p \ge 59$, and this conclusion is the integer case of the fact cited above. The statement $\mathrm{GEN}_{57}$ holds for it with equality in the mass bound, but $\mathrm{GEN}_{56}$ asks for $p \ge 113$ because $S > 56n$. We do not know whether such a graph would have $p \le 112$. If it did, it would refute $\mathrm{GEN}_{56}$ while satisfying REAL-133. Its existence would therefore not by itself separate the two statements, but it shows that a proof of every $\mathrm{GEN}_K$ is a different assertion from REAL-133.

## 9 | Negative results

The examples in this section are explicit graphs.

*Example 9.1.* Let $s \ge 3$, $t \ge 3$ and $m \ge 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of $t$ edges at every cycle vertex, together with $m$ internally disjoint paths $c_0xzyc_1$. Then $n = 2s(t+1) + 3m$, $e = 2s(t+1) + 4m$ and $T = 0$, so $e - n - T = m \ge 1$. For $(s,t,m) = (3,3,1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. They also show that $e = n + T$ fails for radius-extremal strong $C_4$-free graphs of every radius $r \ge 6$: here $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s+t) - 1$.

Two vertices have at most one common neighbour, so $G$ is strong $C_4$-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and $c_0$ remains within distance two of every vertex of such a path, so the eccentricity of $c_0$ is still $s+t$. A vertex $w$ of an added path meets the tip of the arm at $c_s$ only through $c_0$ or $c_1$, hence at distance at least $s+t$. Thus $r = s+t$. For the path order, let $B$ be the block formed by $C$ and the added paths. An induced path $P$ meets $B$ in a subpath $Q$ and can leave $B$ only at an end of $Q$ that is a cycle vertex, through at most $t$ arm vertices. The same case division as for a single ear shows $|P| \le 2s+2t-1$, using $t \ge 3$ and $s \ge 3$, and the cycle minus one vertex together with the arms at its two ends attains this order.

The last sentence of Example 9.1 abbreviates a case division which we now record. If $Q$ avoids both $c_0$ and $c_1$, then $|Q| \le 2s-2$. If $Q$ contains exactly one of them, then either $Q$ lies on the cycle with $|Q| \le 2s-1$, or $|Q| \le 2s+2$ with at most one cycle end, or $|Q| \le 7$ with no cycle end. If $Q$ contains both, the same three shapes occur with $|Q| \le 2s-1$, $|Q| \le 2s+2$ and $|Q| \le 6$. Adding at most $t$ arm vertices at each cycle end of $Q$ gives $|P| \le \max\{2s-1+2t,\ 2s+2+t,\ 7\} = 2s+2t-1$.

*Example 9.2.* Forbidding induced paths on eleven vertices and requiring $L(v) \ge 6$ at every vertex does not forbid an induced star $K_{1,6}$. Two Petersen graphs sharing a vertex give a triangle-free strong $C_4$-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, $p = 9$ and $\lambda = 6$ at the shared vertex. For $b \ge 2$, gluing $b$ copies of the Petersen graph along a common edge gives a 2-connected graph $B_b$ with $n = 8b+2$, $S = 28b+2$, minimum loss 6, $p = 9$ and $\lambda = 2b+1$ at the two hubs. Thus singleton losses do not bound $\lambda$. These graphs are not counterexamples to $\mathrm{GEN}_5$: the eight non-hub vertices of one Petersen block have deletion loss $28 < 5 \cdot 8 + 1$.

For an oriented edge $(u,v)$ of a connected strong $C_4$-free graph $G$, let $q$ be the maximum order of an induced path whose first two vertices are $u, v$ in that order, and let $E$ be the set of terminal vertices of the induced paths of that order. The endpoint inequality $\mathrm{EP}(q)$ asserts $D_G(E) \le q|E|$ for every such oriented edge of maximum order $q$. For a root $s$ with $p_s(G) = q$, if $E$ is the set of terminal vertices of the induced paths of order $q$ starting at $s$, the rooted inequality $\mathrm{EP}^+(q)$ asserts $D_G(E) \le (q-1)|E|$.

**Proposition 9.3.** *For every integer $q \ge 4$, $\mathrm{EP}(q)$ holds for all connected strong $C_4$-free graphs if and only if $\mathrm{EP}^+(q-1)$ does.*

*Proof.* Both graphs below remain strong $C_4$-free, so $S = 2e - 3T$ and the deletion loss of a vertex set is determined by the edges and triangles that meet it.

First assume $\mathrm{EP}^+(q-1)$, and let $(u,v)$ be an oriented edge of maximum induced-path order $q$, with terminal set $E$. Delete $W = N[u] \setminus \{v\}$, and let $H$ be the component of $v$ in $G - W$. An induced path of $G$ that starts with $uv$ has no further vertex in $N[u]$, because the path is induced, so the remainder is an induced path of $H$ starting at $v$. Conversely, every induced path of $H$ starting at $v$ extends by $u$ to an induced path of $G$, since no vertex of $H$ other than $v$ is adjacent to $u$. Thus $p_v(H) = q-1$, and the terminal set in $H$ is still $E$.

Each $x \in E$ has at most one neighbour in $W$: any such neighbour is adjacent to $u$, and two of them would be two common neighbours of $u$ and $x$. Every neighbour of $x$ outside $H$ lies in $W$. Let $m_W$ be the number of edges between $E$ and $W$, so $m_W \le |E|$, and let $T_{\mathrm{extra}}$ be the number of triangles that meet $E$ and are not contained in $H$. Every such triangle uses one of those edges. Comparing edges and triangles,

$$D_G(E) = D_H(E) + 2m_W - 3T_{\mathrm{extra}} \le D_H(E) + 2|E|.$$

The rooted inequality in $H$ is $D_H(E) \le (q-2)|E|$, because $\mathrm{EP}^+(q-1)$ contributes the factor $(q-1)-1$. Therefore $D_G(E) \le q|E|$.

Conversely, assume $\mathrm{EP}(q)$, and let $(H,v)$ be a connected strong $C_4$-free graph with $p_v(H) = q-1$ and terminal set $E$. Form $G$ by adding a vertex $u$ adjacent to $v$ and, for each $x \in E$, a new vertex $a_x$ adjacent exactly to $u$ and to $x$. Every terminal $x$ fails to be adjacent to $v$: an induced path of order $q-1 \ge 3$ from $v$ to $x$ has no chord $vx$. The vertex $a_x$ has degree two. Any cycle through $a_x$ uses both edges $ua_x$ and $a_xx$, and then continues from $x$ to the other neighbour of $u$ on the cycle. That other neighbour is $v$ or some $a_y$. In the first subcase the $x$–$v$ path in $H$ has length at least two, so the cycle has length at least five. In the second the cycle has the form $u, a_x, x, \dots, y, a_y$ and again has length at least five. A cycle through $u$ that avoids every $a_x$ would need two neighbours of $u$ in $H$, but the only neighbour of $u$ in $H$ is $v$. Thus $G$ is strong $C_4$-free. No new triangle appears: the two neighbours of $a_x$ are nonadjacent, and the neighbours of $u$ are pairwise nonadjacent.

An induced path that starts with $uv$ contains no $a_x$. Otherwise the edge $ua_x$ would be a chord, since $a_x$ is not the successor of $u$ on that path. The induced $uv$-paths are therefore the paths $u$ followed by an induced path of $H$ starting at $v$. Their maximum order is $q$, and their terminal set is $E$. Exactly $|E|$ new edges meet $E$, namely the edges $a_xx$, and no new triangle meets $E$. Hence $D_G(E) = D_H(E) + 2|E|$. The oriented inequality gives $D_G(E) \le q|E|$, so $D_H(E) \le (q-2)|E|$, which is $\mathrm{EP}^+(q-1)$. □

*Example 9.4.* EP(9) fails. Let $G$ have vertices $0, \dots, 14$ and adjacency lists

$$\begin{array}{llll} 0\colon 4,8,11 & 1\colon 5,8,10 & 2\colon 6,8,9 & 3\colon 7,10 \\ 4\colon 0,9,10 & 5\colon 1,9,11 & 6\colon 2,10,11 & 7\colon 3,11 \\ 8\colon 0,1,2,12 & 9\colon 2,4,5 & 10\colon 1,3,4,6,13 & 11\colon 0,5,6,7,14 \\ 12\colon 8,13,14 & 13\colon 10,12 & 14\colon 11,12. & \end{array}$$

The graph is triangle-free and strong $C_4$-free, with $n = 15$, $e = 23$ and $S = 46$. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by

$$12, 8, 2, 9, 4, 10, 3, 7, 11 \qquad \text{and} \qquad 12, 8, 2, 9, 5, 11, 7, 3, 10.$$

Thus $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. No induced path that starts with $12, 8$ has order greater than 9. The neighbours of 8 other than 12 are $0, 1, 2$. A continuation is legal only when the new vertex meets the current end and meets no earlier vertex of the path. Under that rule every maximal induced path is one of the following, and the only two of order 9 are the paths already displayed.

| maximal induced path starting $12, 8$ | order |
|---|---|
| $12, 8, 0, 4, 9, 5$ | 6 |
| $12, 8, 0, 4, 10, 3, 7$ | 7 |
| $12, 8, 0, 4, 10, 6$ | 6 |
| $12, 8, 0, 11, 5, 9$ | 6 |
| $12, 8, 0, 11, 6, 10, 3$ | 7 |
| $12, 8, 0, 11, 7, 3, 10$ | 7 |
| $12, 8, 1, 5, 9, 4$ | 6 |
| $12, 8, 1, 5, 11, 6$ | 6 |
| $12, 8, 1, 5, 11, 7, 3$ | 7 |
| $12, 8, 1, 10, 3, 7, 11$ | 7 |
| $12, 8, 1, 10, 4, 9$ | 6 |
| $12, 8, 1, 10, 6, 11, 7$ | 7 |
| $12, 8, 2, 6, 10, 3, 7$ | 7 |
| $12, 8, 2, 6, 10, 4$ | 6 |
| $12, 8, 2, 6, 11, 5$ | 6 |
| $12, 8, 2, 6, 11, 7, 3$ | 7 |
| $12, 8, 2, 9, 4, 10, 3, 7, 11$ | 9 |
| $12, 8, 2, 9, 5, 11, 7, 3, 10$ | 9 |

The first of the two paths of order 9 ends at 11. The neighbours of 11 are $0, 5, 6, 7, 14$: the vertex 7 is the predecessor, 0 meets 8, 5 meets 9, 6 meets both 2 and 10, and 14 meets 12. The second path ends at 10. The neighbours of 10 are $1, 3, 4, 6, 13$: the vertex 3 is the predecessor, 1 meets 8, 4 meets 9, 6 meets both 2 and 11, and 13 meets 12. Neither path extends. The same computation gives a failure of $\mathrm{EP}^{+}(8)$ on the subgraph induced by $0, \dots, 11$, rooted at 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. The inequalities EP(8) and $\mathrm{EP}^{+}(6)$ are not settled here.

*Example 9.5.* Let $k = p - r$. Call an induced subgraph $H$ of $G$ an *eligible core* if it is proper, connected and dominating, and $p(H) \le p(G) - 2$. The budget CB asks for an eligible core with

$$S(G) - S(H) \le k\big(n - |H|\big) + |H|.$$

Together with REAL-133 for $H$ and an induction on the order, the budget would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$ and $k = 3$, hence $\ell = 3$. An eligible core has $p(H) \le 3$. A connected triangle-free strong $C_4$-free $P_4$-free graph is a star, by Lemma 2.7 and the absence of triangles. In the Petersen graph a star with $k \le 3$ leaves has centre of degree 3. The vertices it dominates are the centre, its three neighbours, and at most two further neighbours of each leaf, at most $4 + 2k$ vertices in all. This is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$, and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$.
The unrestricted budget therefore fails. The example has $\ell = 3$. It does not determine a cutoff such as $\ell > 5$, and this paper contains no census of eligible cores.

*Example 9.6.* The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, and $Z = 28$ nonadjacent pairs with no common neighbour) satisfy the counting identities used above, including $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$, together with the lower bounds $p \ge 2r - 1$ and $p \ge \min_v \lambda(v) + 2$, if $p$ is replaced by the value 5. But $5 < r + \ell = 6$. This system of scalar constraints does not imply REAL-133. The Heawood graph itself has $p = 7$.

*Example 9.7.* Let $F$ be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2+q+1)$ and $S(F) = (q+1)f$, and choose a root of $F$. Let $t = f - 1$ and $b \ge 1$. Take a 6-cycle $c_0 \cdots c_5$, attach a pendant path with $t$ edges at each cycle

vertex, and attach $b$ copies of $F$ at $c_0$ by identifying their roots with $c_0$. The result is strong $C_4$-free, because each new block meets the rest in a cut vertex, and

$$n = 6f + b(f-1), \qquad S = 12f + b(q+1)f, \qquad r = t+3, \qquad p = 2t+5 = 2r-1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f-1 \le t$ of $c_0$, so each cycle vertex has eccentricity $t+3$, while a vertex $x$ of a copy has distance $d_F(x,c_0)+t+3$ from the tip of the arm at $c_3$. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most $t$ vertices. Mass is additive at cut vertices by Lemma 2.3. For fixed $q$, as $b \to \infty$ one has $\ell \to (q+1)f/(f-1)$. Letting $q \to \infty$, with $r = f+2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy RAD($r$) for their own radius, and they show that $\ell$ is unbounded on the class $p = 2r-1$. Any theorem RAD($r$) must use a bound that depends on $r$.

## 10 | What is proved and what is open

The comparisons with Chen [3], Chen and Xu [5], Chen, Wu and Xu [4], Chudnovsky and Stacho [6], DeLaViña, Larson, Pepper and Waller [8], Karthick and Maffray [12], and Brandstädt and Hoàng [1] are in the introduction, in Section 5 and in Remarks 3.7 and 3.8. Table 1 is the only status list. An open row is not a hypothesis of a proved row. A finite search is not a proof here.

Suppose $4 < \ell \le 5$ and $G$ violates REAL-133. Then $\lceil \ell \rceil = 5$, so Corollary 6.1 with $K = 4$ gives either $r \le 4$ and $2r-1 \le p \le r+4$, or $r = 5$ and $p = 9$. Lemma 5.7 forces $\ell < 4$ whenever $p \le 6$, so $p \ge 7$. Corollary 4.4 forces $\ell \le 2$ when $(r,p) = (4,7)$. The pairs that remain are $(3,7)$, $(4,8)$ and $(5,9)$.

For $5 < \ell \le 6$, Corollary 6.1 gives $2r-1 \le p \le r+5$ and $r \le 6$. Lemma 3.1 excludes radius 1, Theorem 3.6 excludes $p \le 5$, and Corollary 4.4 excludes $(4,7)$. Lemma 5.7 excludes $(2,6)$ and $(3,6)$. Lemma 6.4 shows that a triangle-free graph with $p = 7$ has $\ell < 4$, so the pairs $(2,7)$ and $(3,7)$ are possible in this range only for graphs that contain a triangle. Lemma 6.3 is the stronger bipartite exclusion of $(2,7)$. The pairs that remain are necessary conditions for a counterexample. The table does not assert that any of them occurs.

| Statement | Status |
|---|---|
| $M(n)$ for connected strong $C_4$-free $P_6$-free graphs | proved, Theorem 3.6 |
| the structure of graphs with $r = 4$ and $p = 7$, the formula $S = 2n-b-m$, and the equality family | proved, Theorem 4.3 and Corollaries 4.4, 4.5 |
| $2e \le 3n$ for connected triangle-free strong $C_4$-free $P_7$-free graphs, equality only for the Petersen graph | proved, Theorem 5.4 |
| $\Delta \ge 1$ for $n \ge 4$, hence $e \le 2n-4$ and $\ell < 4$; if $T \ge 1$ and $n \ge 5$, then $S \le 4n-13$ | proved, Lemmas 5.7 and 5.8 |
| REAL-133 for $\ell \le 3$; for $r \le 2$ and $\ell \le 4$; for $r = 4$ and $\ell \le 4$; for $r \ge 5$ and $\ell \le 4$ | proved, Theorem 5.11 |
| ROOT(3) and ROOT(4); the terminal strengthening of Proposition 7.3 | proved |
| ROOT($q$) for every $q$, in the class $\mathcal{B}$ | proved, Theorem 8.2 |
| a minimum counterexample to $\mathrm{GEN}_4$ is 2-connected | proved, Proposition 7.4; $\mathrm{GEN}_4$ is open |
| the cell $r = 3$, $p = 6$, $3 < \ell \le 4 - 13/n$, $n \ge 14$, $T \ge 1$ | open; Corollary 5.9, and $\ell = 4$ does not occur |
| $\mathrm{GEN}_3$ for graphs that contain a triangle | open; Appendix A is conditional |
| $\mathrm{GEN}_K$ for $K \ge 4$, unrestricted $\mathrm{GEN}_6$, RAD($r$) for $r \ge 5$, and ROOT($q$) for $q \ge 5$ outside $\mathcal{B}$ | open |
| REAL-133 | open |
| pairs $(r,p)$ still necessary for $4 < \ell \le 5$ | $(3,7)$, $(4,8)$, $(5,9)$ |
| pairs $(r,p)$ still necessary for $5 < \ell \le 6$ | $(2,7)$, $(3,7)$, $(3,8)$, $(4,8)$, $(4,9)$, $(5,9)$, $(5,10)$, $(6,11)$ |
| pairs excluded from $5 < \ell \le 6$ for every strong $C_4$-free graph | $(2,6)$ and $(3,6)$, by Lemma 5.7 |
| $(2,7)$ and $(3,7)$ in the range $5 < \ell \le 6$ | each such graph contains a triangle, by Lemma 6.4 |

**TABLE 1** | Proved statements and statements left open. The rows for $4 < \ell \le 5$ and for $5 < \ell \le 6$ record necessary pairs only.

No structure theorem for connected strong $C_4$-free graphs of radius $r \ge 5$ and path order $2r-1$ is proved here. Notes on deletion in the triangle-free class, and on ports of a distance-preserving dominating set, are kept outside this article in `notes/supplement-deletion-ports.tex`.

**Conjecture 10.1.** ROOT($q$) *holds for every $q \geq 3$. For every $r \geq 5$, every connected strong $C_4$-free graph of radius $r$ and path order $2r-1$ satisfies $S \leq (r-1)n - 2r(r-3)$, with equality only for $C_{2r}$.*

The cases $q = 3$ and $q = 4$ are Lemma 5.3 and Theorem 7.2, and every $q$ holds in $\mathcal{B}$ by Theorem 8.2. The radius bound is not claimed for $r \geq 5$.

**Conjecture 10.2.** *If $F$ is connected and strong $C_4$-free, $p_s(F) = q$ and the eccentricity of $s$ is $q-1$, then*

$$S(F) \leq (q-1)(n-1) - (q-3)D - m - (q-2)(q-3).$$

For $q = 4$ the subtracted term is 2, and Proposition 7.3 is this case. For $q \geq 5$ the conjecture is open. The terminal bound in $\mathcal{B}$ is the first part of Theorem 8.2.
Conjecture 10.2, for a fixed $q$, implies ROOT($q$). The implication does not prove the conjecture. Let $F$ be connected and strong $C_4$-free with $p_s(F) \leq q$. Form $G$ by attaching a path of $q-1$ new edges at $s$. Then $G$ is terminal of index $q$: the new path is an induced path of order $q$ starting at $s$, and every induced path of $F$ that starts at $s$ has order at most $q$. The order, the root degree and the mass increase by $q-1$, 1 and $2(q-1)$ respectively, and $m$ is unchanged. Write $N = n+q-1$ and $D_G = D+1$. The terminal inequality in $G$ reads

$$S(F) + 2(q-1) \leq (q-1)(N-1) - (q-3)D_G - m - (q-2)(q-3).$$

The identity $N - 1 = n + q - 2$ gives $(q-1)(N-1) = (q-1)(n-1) + (q-1)^2$, and $(q-3)D_G = (q-3)D + (q-3)$. The terms that do not involve $n$, $D$ or $m$ are

$$(q-1)^2 - (q-3) - (q-2)(q-3) - 2(q-1) = (q-3)\big((q-1) - (q-1)\big) = 0.$$

What remains is $S(F) \leq (q-1)(n-1) - (q-3)D - m$, which is ROOT($q$). For $q = 4$ this route uses Proposition 7.3. The induction in Theorem 7.2 does not. For $q \geq 5$ the hypothesis is open, so the implication leaves ROOT($q$) open.

**Conjecture 10.3.** *Every connected strong $C_4$-free graph $G$ that is not a cycle and has $\ell(G) > 5$ has an eligible core $H$, in the sense of Example 9.5, with*

$$S(G) - S(H) \leq (p-r)\big(n - |H|\big) + |H|.$$

Example 9.5 shows that the budget without a restriction on $\ell$ is false, and that the Petersen graph is a counterexample with $\ell = 3$. The example does not show that 5 is the correct cutoff, and it does not count eligible cores. A reported count of cores of order at most 12 is not a result of this paper. If the conjecture, the range $\ell \leq 5$, and an induction on the order were all available, REAL-133 would follow. The range $\ell \leq 5$ is not a theorem, so this route is not a proof of REAL-133. Example 9.6 shows that a proof cannot rely only on the scalar constraints listed there.

## APPENDIX

## A | A conditional reduction for $\mathrm{GEN}_3$

Theorem 5.4 is the triangle-free case of $\mathrm{GEN}_3$. The three local statements named at the end of this appendix are not proved for graphs that contain triangles. With them, the hand arguments below would give $\mathrm{GEN}_3$ in general. Without them, that case remains open. The cell $r = 3$, $p = 6$, $3 < \ell \leq 4 - 13/n$, of order at least 14 and with at least one triangle, remains open as well. By Corollary 5.9 one has $\ell \neq 4$.

**Theorem A.1.** *Let $G$ be a counterexample to $\mathrm{GEN}_3$ of minimum order: $G$ is connected and strong $C_4$-free, $p(G) \leq 6$, $S(G) \geq 3n+1$, and $n$ is minimum. Then $G$ is 2-connected; $S(G) - S(G-U) \geq 3|U| + 1$ for every nonempty proper $U \subset V(G)$; $2\lambda(v) - t(v) \geq 4$ for every $v$, so $\lambda(v) \geq 2$; and every vertex with $\lambda(v) = 2$ has degree 2 and nonadjacent neighbours, and no two vertices of degree 2 are adjacent.*

*Proof.* Since $S \leq 2 < 3n$ for $n \leq 2$, one has $n \geq 3$. Let $v$ be a cut vertex. For each component $C$ of $G - v$ let $\rho_C$ be the largest order of an induced path in $G[C \cup \{v\}]$ with end $v$. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices, so some component $C_1$ has $\rho_{C_1} \leq 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Both graphs are connected, smaller than $G$, and $P_7$-free. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$, and minimality gives $S(G_2) \leq 3n_2$. By Lemma 2.3, $S(G) \leq 2(n_1-1) + 3n_2 = 3n - (n_1 - 1) < 3n$.
Every component $H$ of $G - U$, for $U$ nonempty and proper, is connected, smaller, and $P_7$-free, so $S(H) \leq 3|H|$. Summing, $S(G-U) \leq 3(n-|U|)$, and $S(G) \geq 3n+1$ gives the deletion bound. The case $U = \{v\}$ and Lemma 2.2 give $2\lambda(v) - t(v) \geq 4$. If $\lambda(v) = 2$, then $t(v) = 0$, so $d(v) = 2$ and the neighbours are nonadjacent. Every degree-2 vertex has $\lambda = 2$ and hence

lies in no triangle. If $u \sim w$ both have degree 2, with other neighbours $x$ and $y$, then $x \neq y$ because $t(u) = 0$. The set $\{u\}$ is a unit of $x$ and $\{w\}$ is a unit of $y$, so Lemma 2.2 with $U = \{u, w\}$ gives loss $2 + 2 + 1 + 1 = 6 < 7$. □

The same mass bound for a disconnected graph is the sum of the bounds on its components. An isolated vertex has $S = 0$. Call a graph *admissible* if it is strong $C_4$-free and $P_7$-free, $L(v) \geq 4$ for every vertex, and no two adjacent vertices have degree 2. A minimum counterexample is admissible. In an admissible graph, $\lambda(v) = 2$ forces $t(v) = 0$ and $d(v) = 2$. Write $B$ for the vertices of degree at least 3, the *branch vertices*, and $Z$ for the degree-2 vertices, the *ears*. Put $D = G[B]$, $d_D(v) = |N(v) \cap B|$ and $q(v) = |N(v) \cap Z|$. All triangles lie in $D$.

**Lemma A.2.** *In an admissible graph, $q(v) \leq 4$ for every vertex $v$.*

*Proof.* Let $u_i$ be ears at $v$ with other neighbours $a_i$. The $a_i$ are distinct, nonadjacent to $v$, and of degree at least 3. Put $E_i = N(a_i) \setminus \{u_i\}$. Then $|E_i| \geq 2$ and $E_i \cap N[v] = \emptyset$.
If $a_i \nsim a_j$, then $E_i = \{w, p\}$ and $E_j = \{w, q\}$ with $pq \in E$, $w \nsim p$ and $w \nsim q$. Indeed two common neighbours of $a_i$ and $a_j$ are forbidden, so $|E_i \cap E_j| \leq 1$. Since $|E_i| \geq 2$, the private part $E_i \setminus E_j$ is nonempty, and likewise $E_j \setminus E_i$. If $p$ lies in the first private part and $q$ in the second with $p \nsim q$, the path $pa_iu_iva_jq$ is induced. Thus the private parts are complete to each other. If one of them contained two vertices, those two vertices and one vertex of the other private part would form a 4-cycle with the corresponding $a$-vertex. Both private parts therefore have size one, and $|E_i \cap E_j| = 1$. An edge from the common vertex to a private vertex is another 4-cycle. In particular $a_i$ and $a_j$ have degree 3 and lie in no triangle.
No three of the vertices $a_i$ are pairwise nonadjacent. Otherwise the sets $E_i$ meet pairwise in a single vertex. If the three intersections are the three pairs of a triple, the claim forces a triangle on that triple, and one $a_i$ together with two vertices of the triple forms a 4-cycle. Otherwise $E_i = \{w, p_i\}$ with $p_1p_2p_3$ a triangle and $w$ nonadjacent to each $p_i$, and $vu_1a_1wa_2p_2p_3$ is an induced $P_7$.
Five ears are impossible. The graph $F$ they determine on $\{a_1, \dots, a_5\}$ has independence number at most two. If some vertex $a$ of $F$ is adjacent to the other four, those four induce a matching. The independence number forces that matching to be perfect, say $\{b_1b_2, c_1c_2\}$. Then $b_1 \nsim c_1$, so the preceding paragraph says that $b_1$ lies in no triangle, but $ab_1b_2$ is a triangle. Otherwise every vertex of $F$ has a non-neighbour, hence has degree 3 in $G$ and lies in no triangle. One neighbour of each such vertex is its ear, so $F$ has maximum degree at most two. It is triangle-free and has no 4-cycle, and its independence number is at most two, so $F$ is a 5-cycle. Label that cycle $a_1a_2a_3a_4a_5a_1$, and write $u_i$ for the ear joining $v$ to $a_i$. The seven vertices $u_5, v, u_1, a_1, a_2, a_3, a_4$, in that order, induce a $P_7$: each $u_i$ has neighbours $v$ and $a_i$ only, $v$ is nonadjacent to every $a_j$, and the only edges of the cycle among $a_1, a_2, a_3, a_4$ are the three path edges. □

**Proposition A.3.** *Let $G$ be admissible, with $B$, $Z$, $D$, $d_D$ and $q$ as above. Suppose that $d_D(v) \leq 3$ for every $v$, that $d_D(v) = 3$ implies $q(v) = 0$, and that $d_D(v) = 2$ with the two neighbours in $B$ nonadjacent implies $q(v) \leq 2$. Then $S(G) \leq 3n$.*

*Proof.* Ears have nonadjacent neighbours and $Z$ is an independent set, so $S(G) = S(D) + 4|Z|$. Since $|Z| = \frac{1}{2}\sum_{v \in B} q(v)$ and $n = |B| + |Z|$,

$$S - 3n = \sum_{v \in B} \Big( d_D(v) - t_D(v) - 3 + \tfrac{1}{2} q(v) \Big).$$

Lemma A.2 gives $q(v) \leq 4$. If $d_D(v) = 2$ and $t_D(v) = 0$, the two neighbours in $B$ are nonadjacent, for an edge between them would be a triangle. The three suppositions and the ear bound make every summand nonpositive: the summand is at most $-1$ when $(d_D, t_D) = (0, 0)$, at most 0 when $(d_D, t_D) = (1, 0)$, at most 0 when $(d_D, t_D) = (2, 0)$, at most 0 when $(d_D, t_D) = (2, 1)$, and equal to $-t_D(v) \leq 0$ when $d_D(v) = 3$. The suppositions exclude $d_D(v) \geq 4$. □

The three suppositions in Proposition A.3 are not proved in this paper. Theorem A.1 shows that a minimum counterexample to $\mathrm{GEN}_3$ is admissible, so those suppositions would imply $\mathrm{GEN}_3$. They do not, by themselves, make $\mathrm{GEN}_3$ a theorem.

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